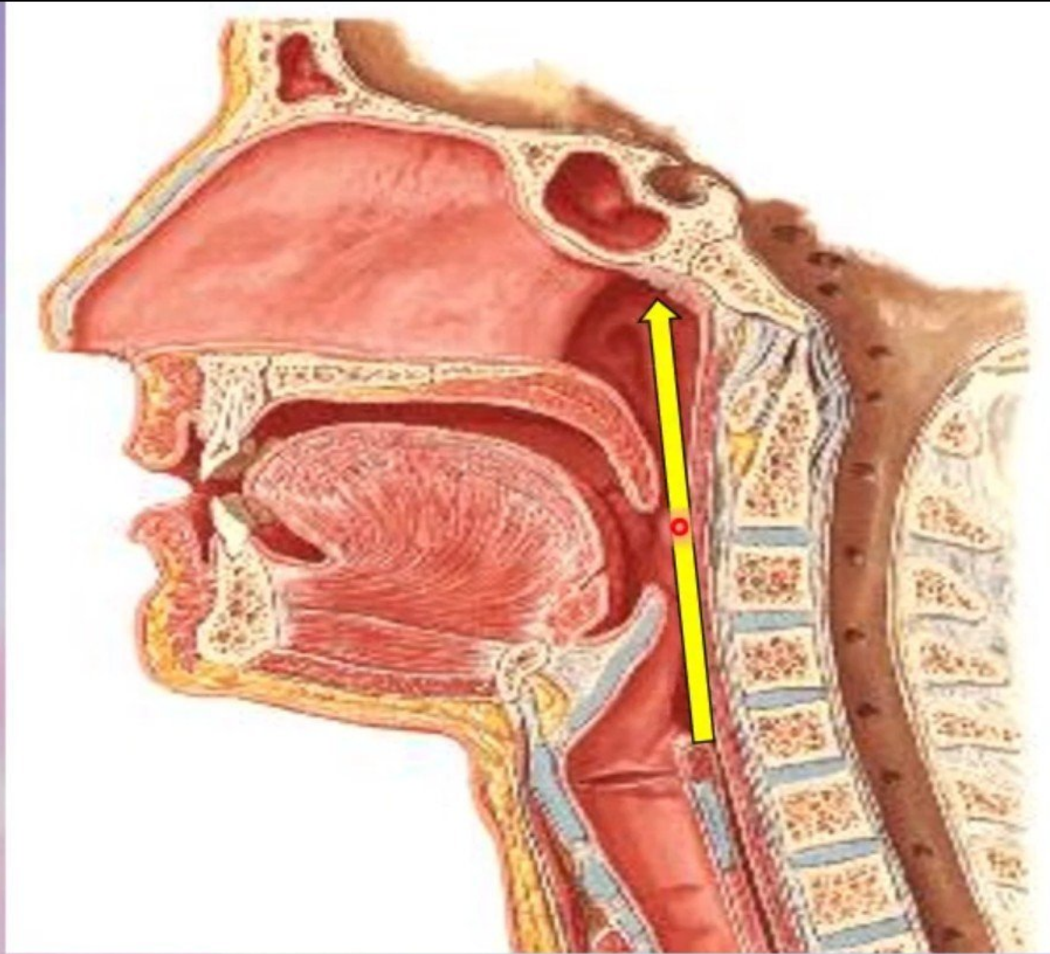
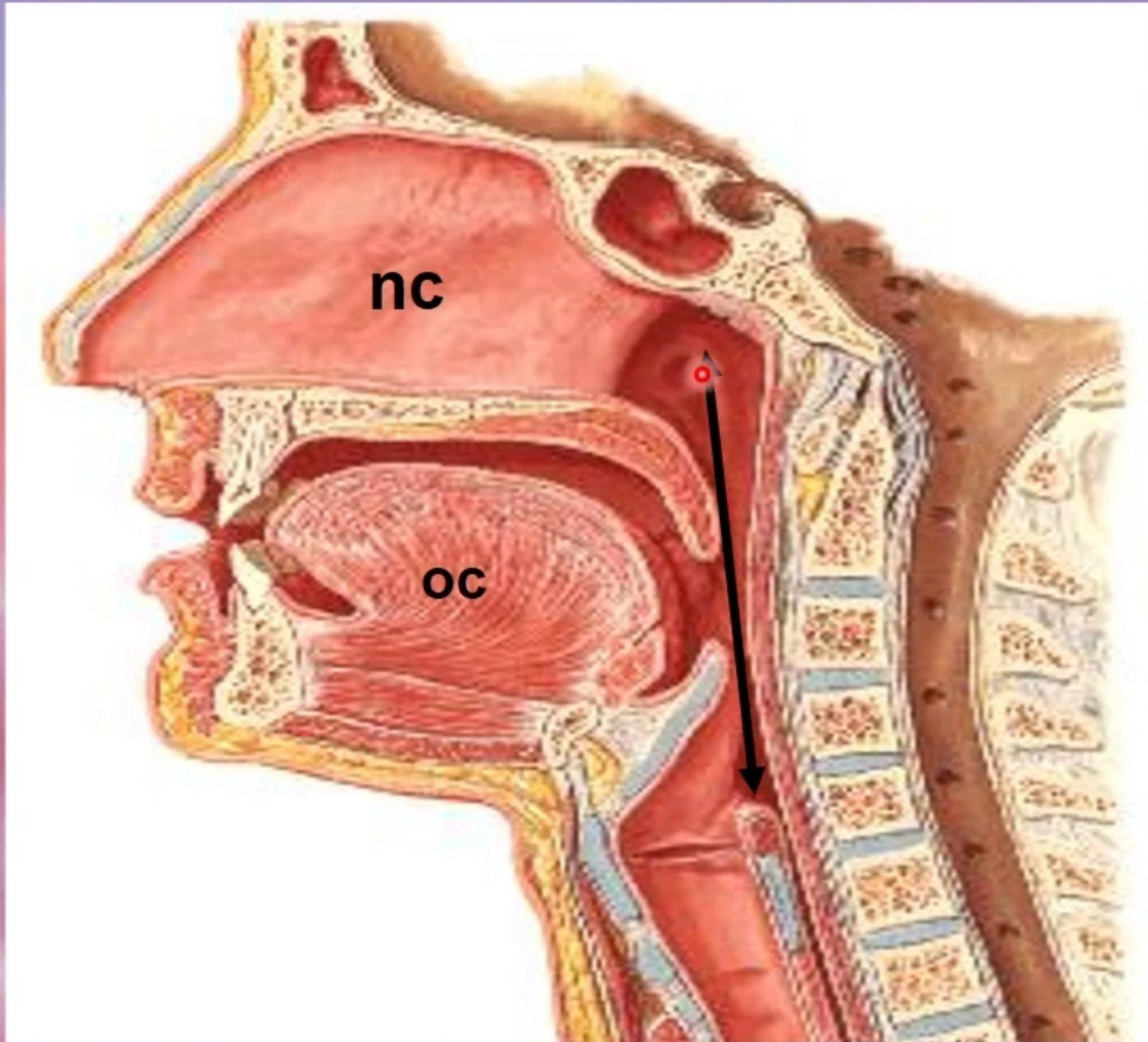
# Pharynx

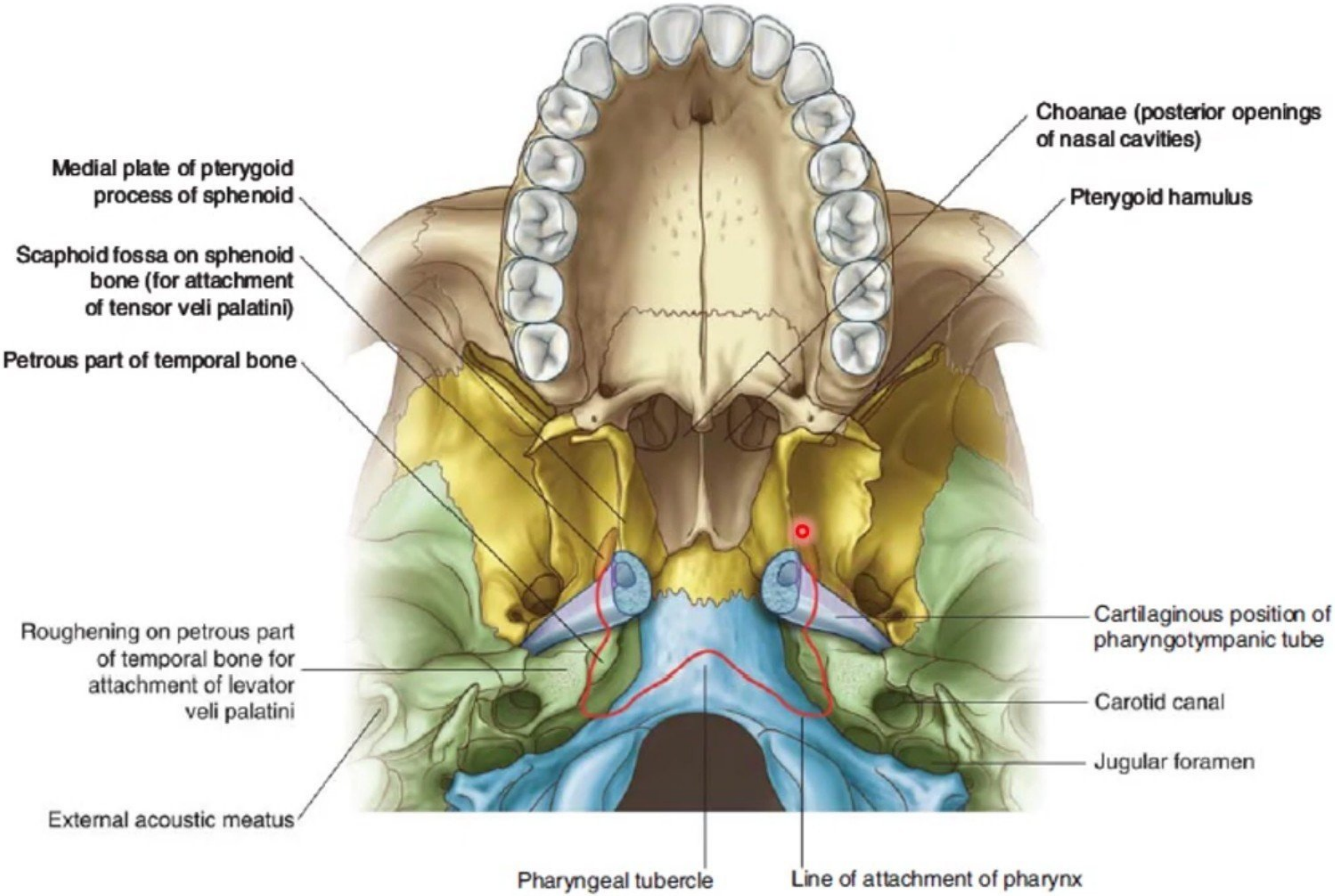


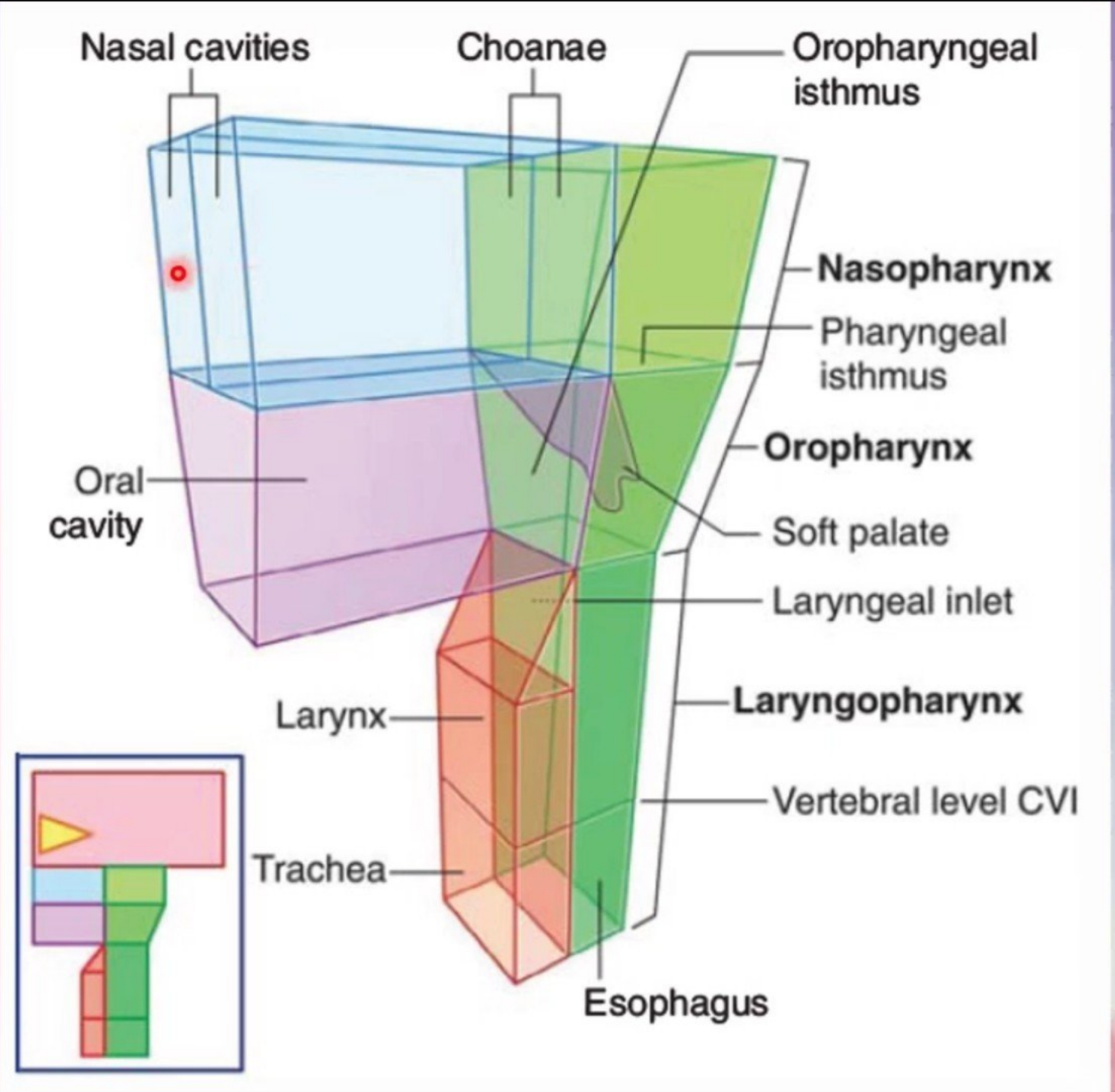
Dr. Osman Ahmed Osman
Department of Anatomy

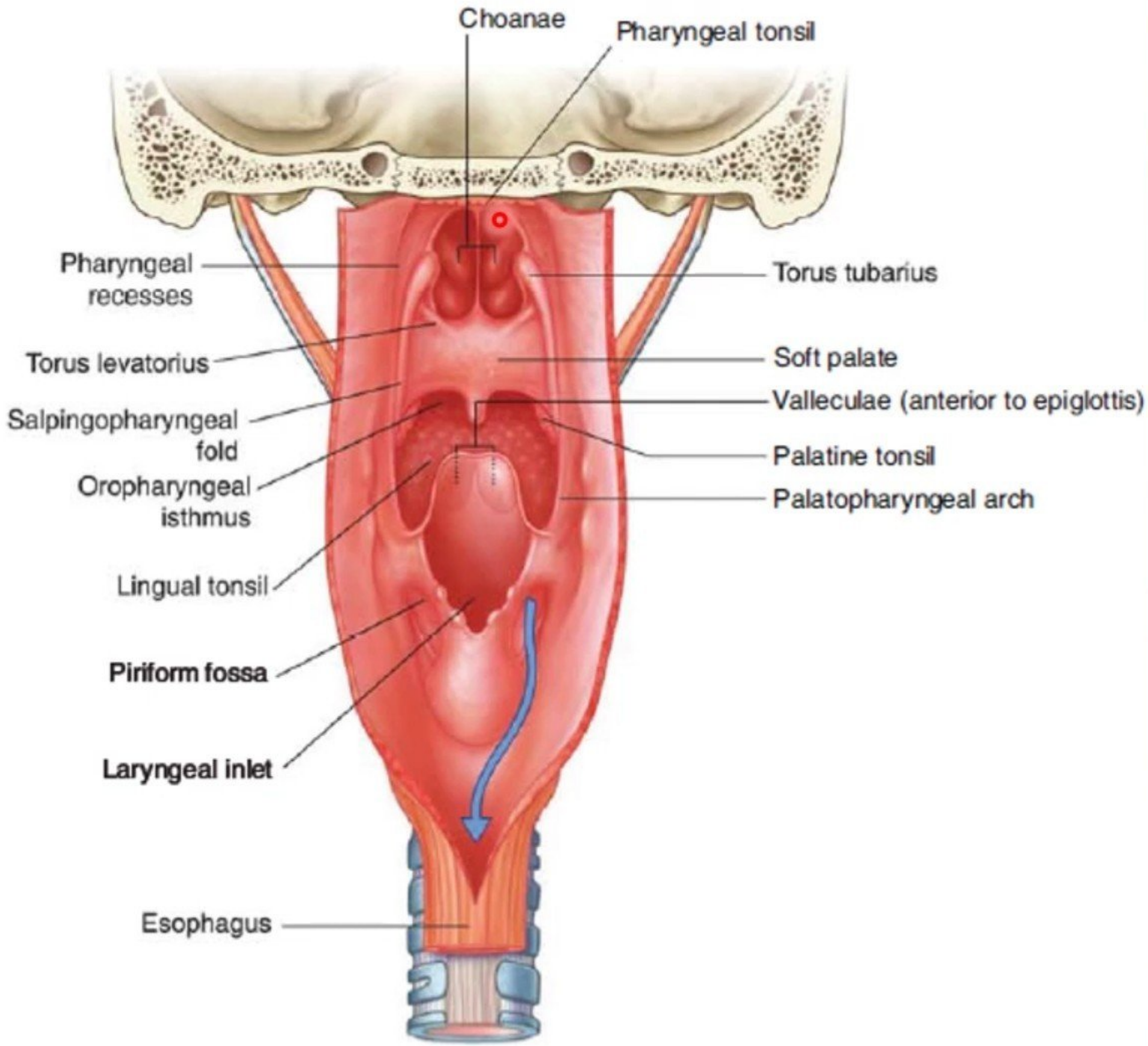
# Pharynx

- Muscular tube lying behind the nose, oral cavity & larynx.
- Extends from the base of the skull to level of the 6th cervical vertebra, where it is continuous with the esophagus
- The anterior wall is deficient and shows (from above downward):
  - Posterior nasal apertures.
  - Opening of the oral cavity.
  - Laryngeal inlet.
- The muscles arranged in circular and longitudinal layers.

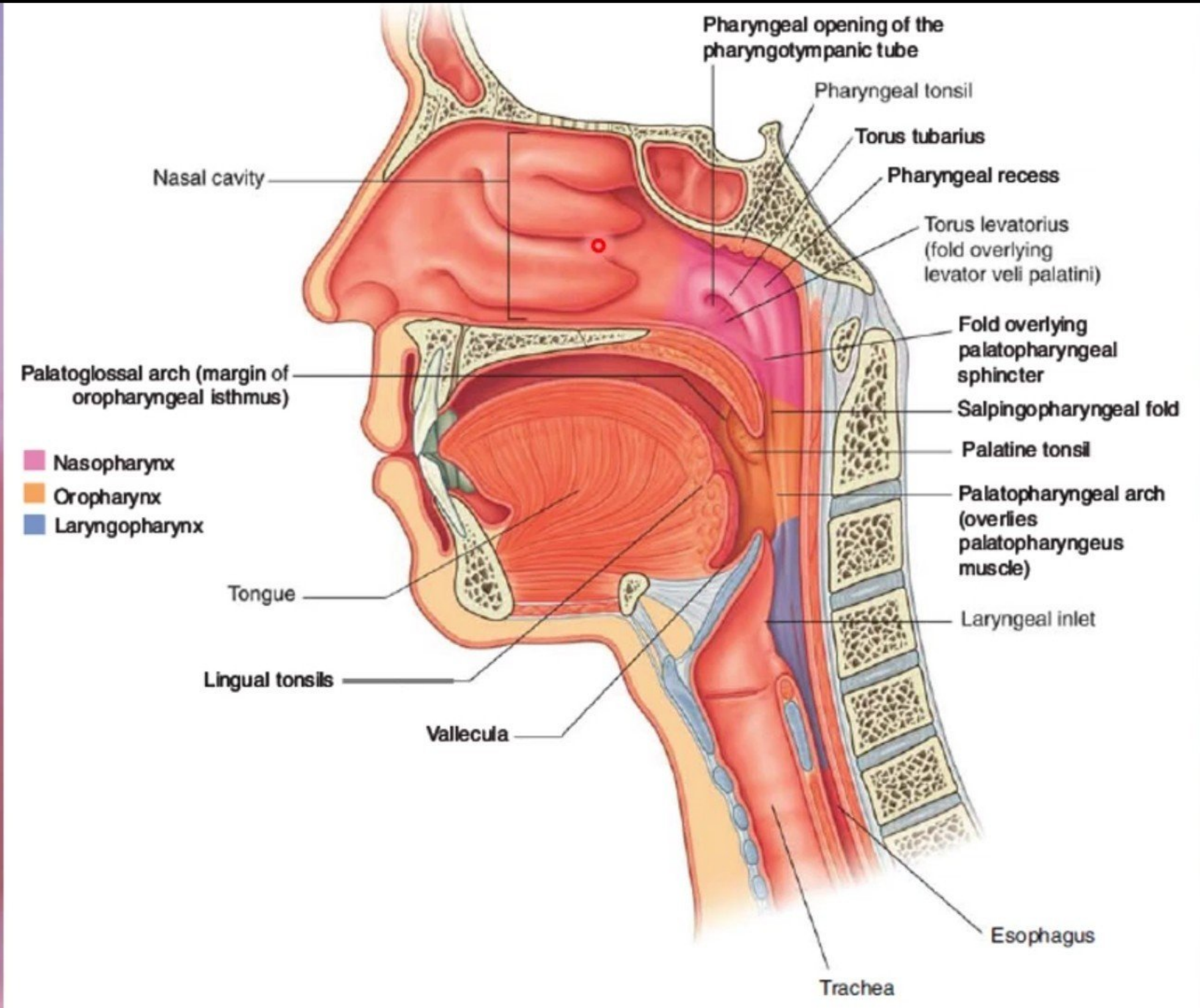


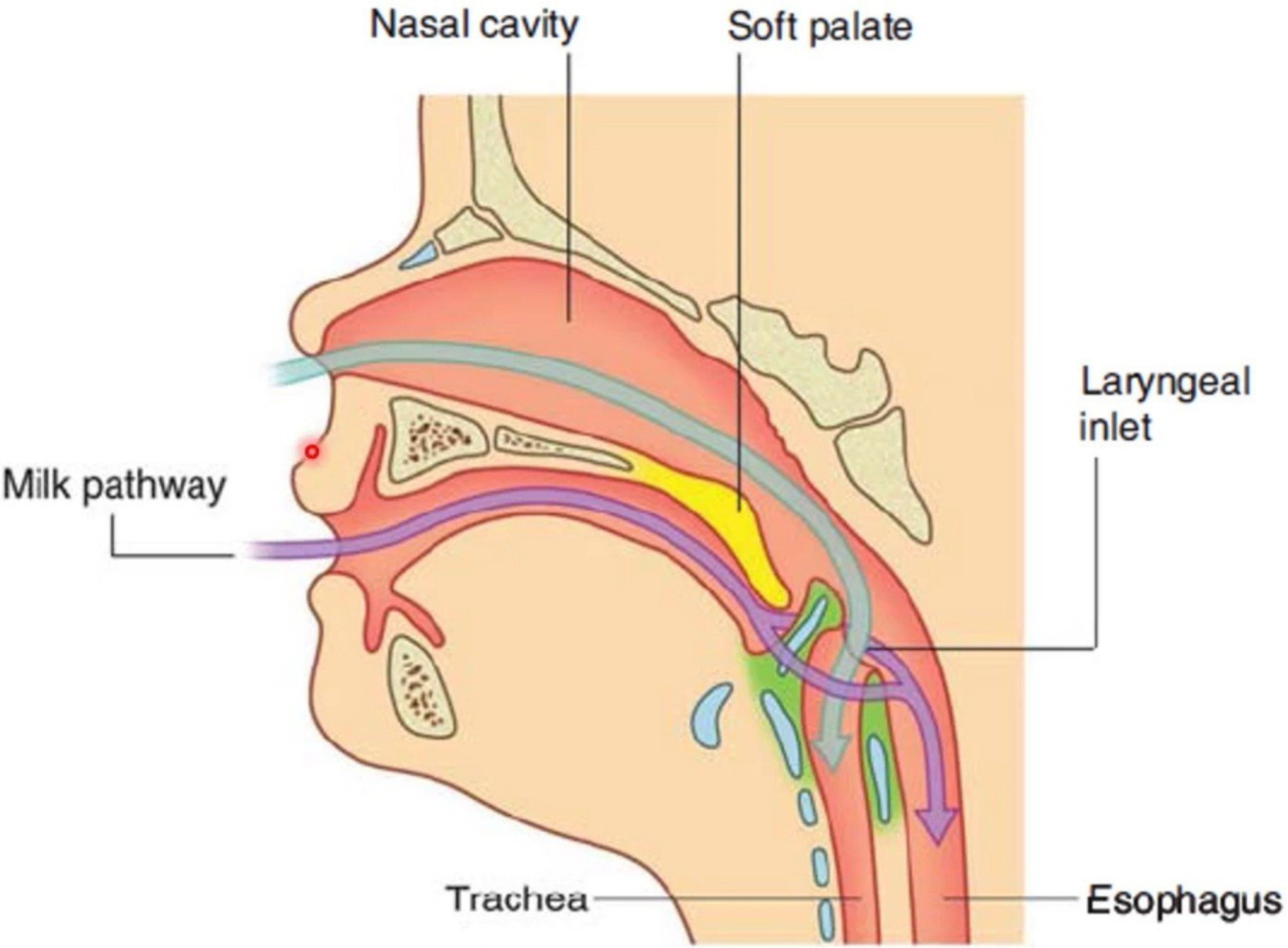

Medial plate of pterygoid process of sphenoid
Scaphoid fossa on sphenoid bone (for attachment of tensor veli palatini)
Petrous part of temporal bone
Roughening on petrous part of temporal bone for attachment of levator veli palatini
External acoustic meatus
Choanae (posterior openings of nasal cavities)
Pterygoid hamulus
Cartilaginous position of pharyngotympanic tube
Carotid canal
Jugular foramen
Pharyngeal tubercle
Line of attachment of pharynx


Nasal cavities
Choanae
Oropharyngeal isthmus
Nasopharynx
Pharyngeal isthmus
Oropharynx
Oral cavity
Soft palate
Laryngeal inlet
Larynx
Laryngopharynx
Vertebral level CVI
Trachea
Esophagus



Choanae
Pharyngeal tonsil
Pharyngeal recesses
Torus tubarius
Torus levatorius
Soft palate
Valleculae (anterior to epiglottis)
Salpingopharyngeal fold
Palatine tonsil
Oropharyngeal isthmus
Palatopharyngeal arch
Lingual tonsil
Piriform fossa
Laryngeal inlet
Esophagus


Pharyngeal opening of the pharyngotympanic tube
Pharyngeal tonsil
Torus tubarius
Pharyngeal recess
Torus levatorius (fold overlying levator veli palatini)
Nasal cavity
Fold overlying palatopharyngeal sphincter
Salpingopharyngeal fold
Palatoglossal arch (margin of oropharyngeal isthmus)
Palatine tonsil
Nasopharynx
Oropharynx
Laryngopharynx
Palatopharyngeal arch (overlies palatopharyngeus muscle)
Tongue
Laryngeal inlet
Lingual tonsils
Vallecula
Esophagus
Trachea

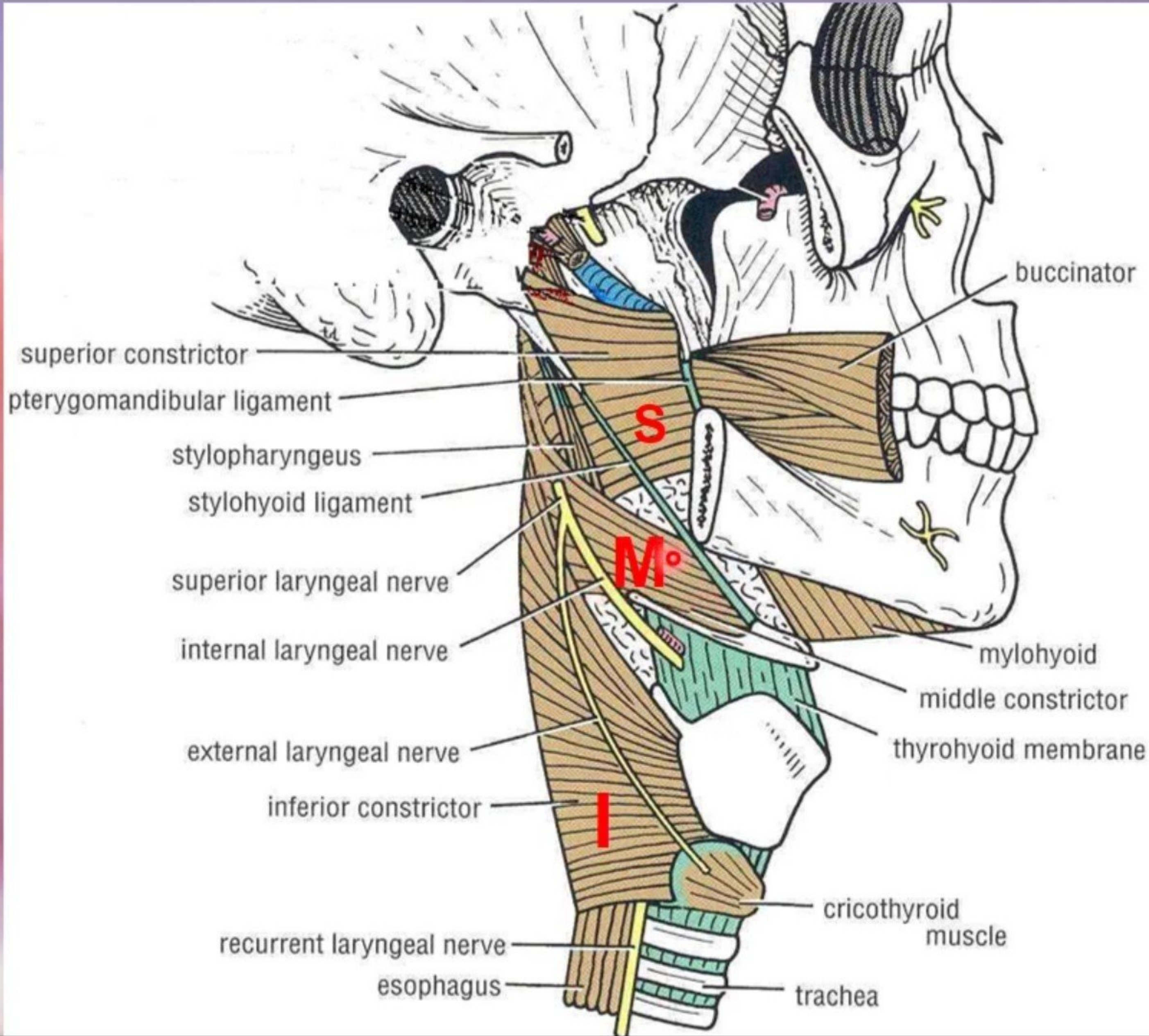

Nasal cavity
Soft palate
Laryngeal
inlet
Milk pathway
Trachea
Esophagus

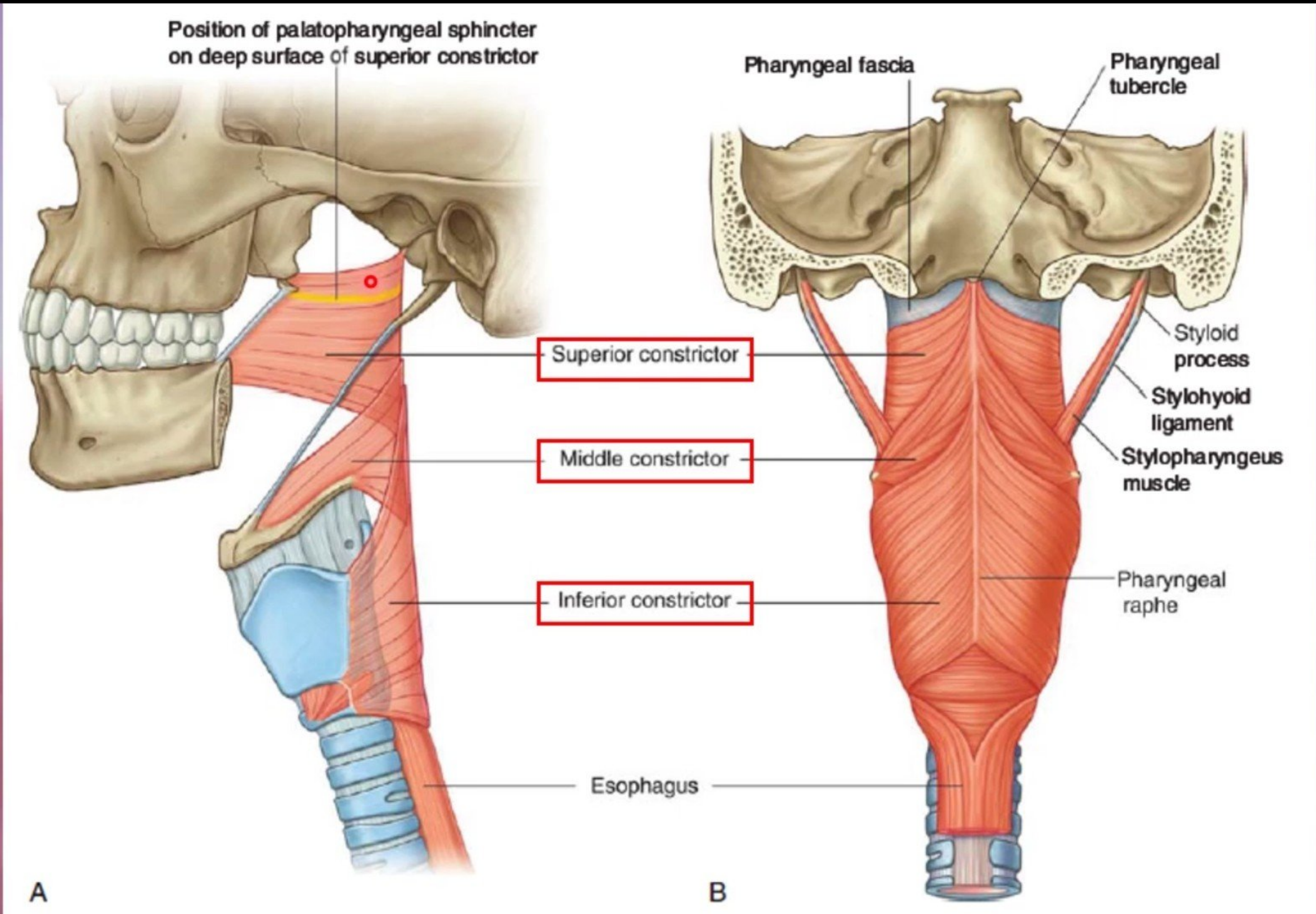
## Circular (Constrictor) Muscles

- Three in number:
  - **Superior constrictor,**
  - **Middle constrictor &**
  - **Inferior constrictor**
- The three muscles overlap each other.

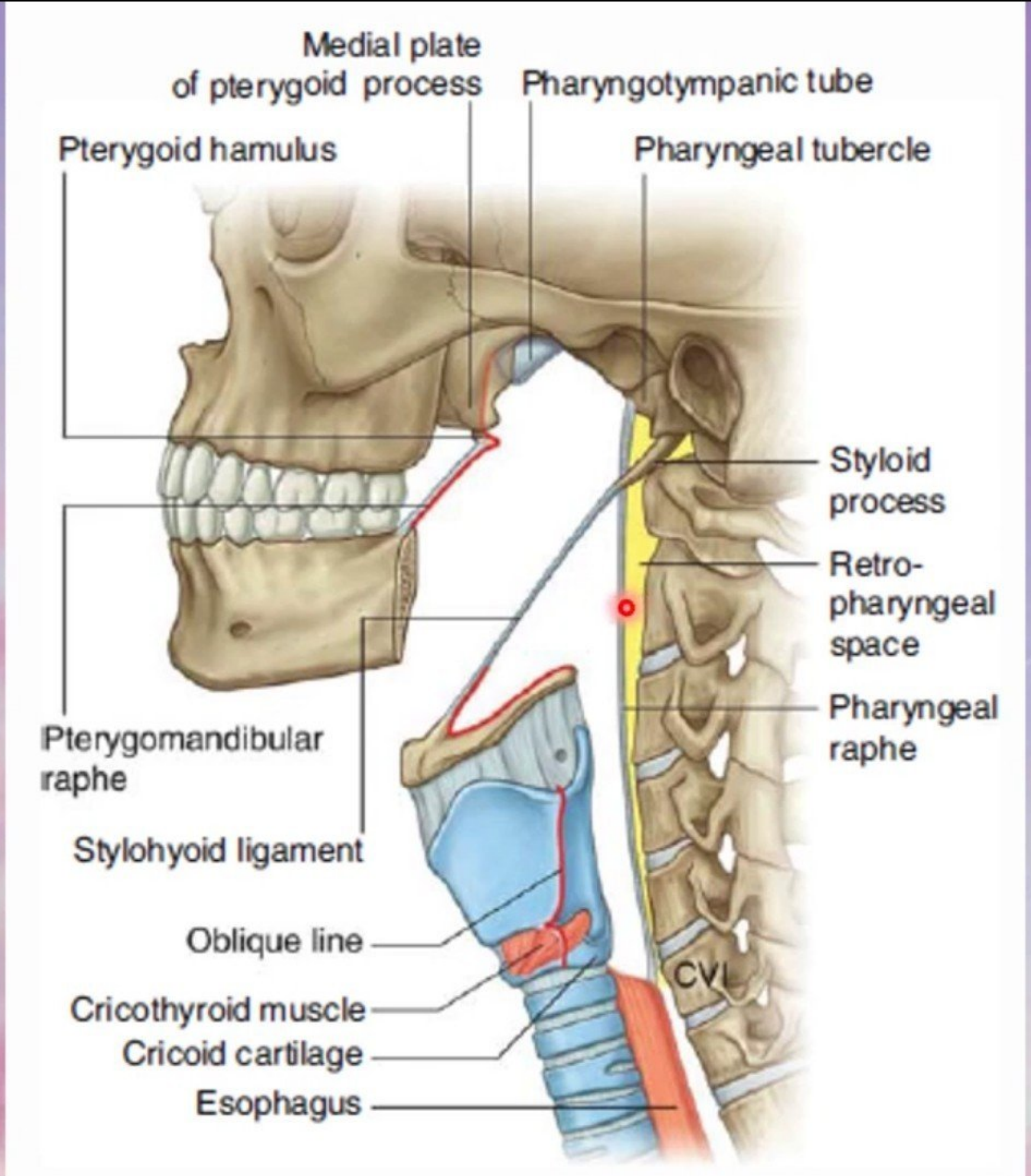
**Functions:**

- Propel the bolus of food down into the esophagus.
- lower fibers of the inferior constrictor (**Cricopharygeus**) act as a sphincter, preventing the entry of air into the esophagus between the acts of swallowing.




Position of palatopharyngeal sphincter
on deep surface of superior constrictor
Superior constrictor
Middle constrictor
Inferior constrictor
Esophagus
A
Pharyngeal fascia
Pharyngeal
tubercle
Styloid
process
Stylohyoid
ligament
Stylopharyngeus
muscle
Pharyngeal
raphe
B

# Attachment of lateral pharyngeal wall



**The gap between the superior and middle constrictors:**

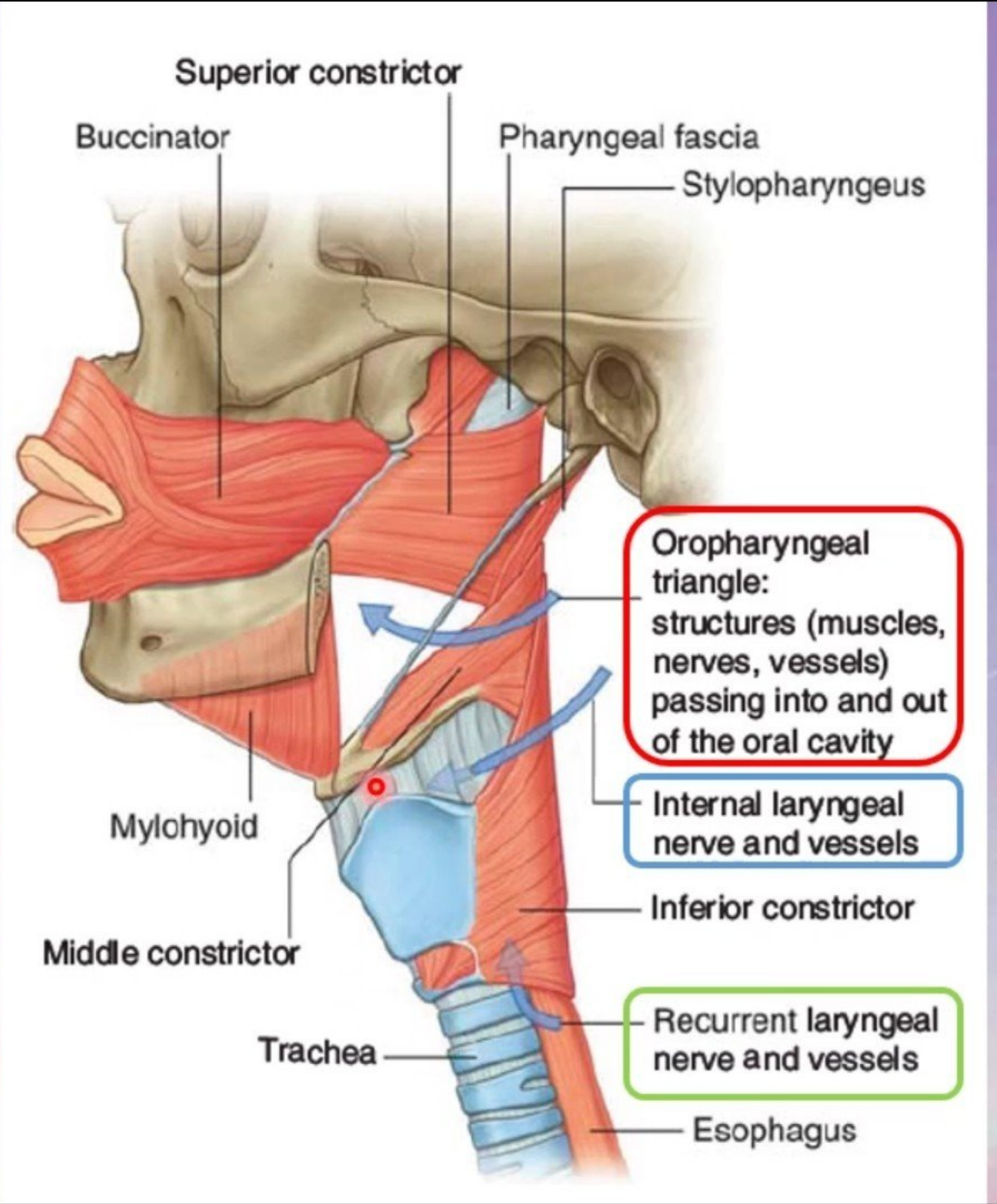
- Stylopharyngeus passes down into the pharynx.
- Styloglossus.
- Glossopharyngeal to the tongue.
- lingual nerves pass to the tongue.

**The anterior gap between the middle and inferior constrictors is closed by the thyrohyoid membrane in this gap by piercing the membrane are:**

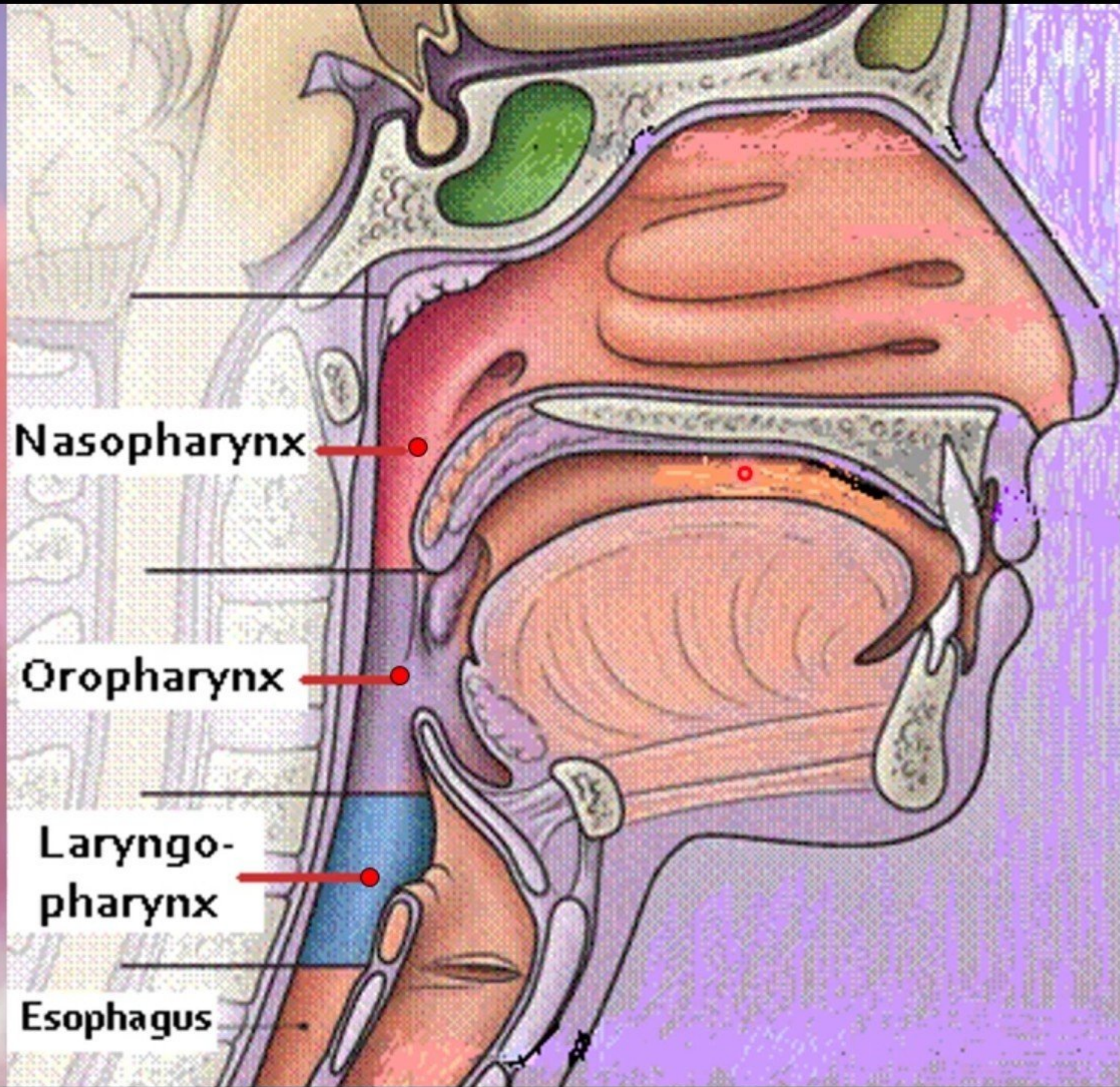
- The internal laryngeal nerve.
- The superior laryngeal vessels.

**The structures passing upwards deep to the lower border of the inferior constrictor are:**

- The recurrent laryngeal nerve.
- The inferior laryngeal vessels.

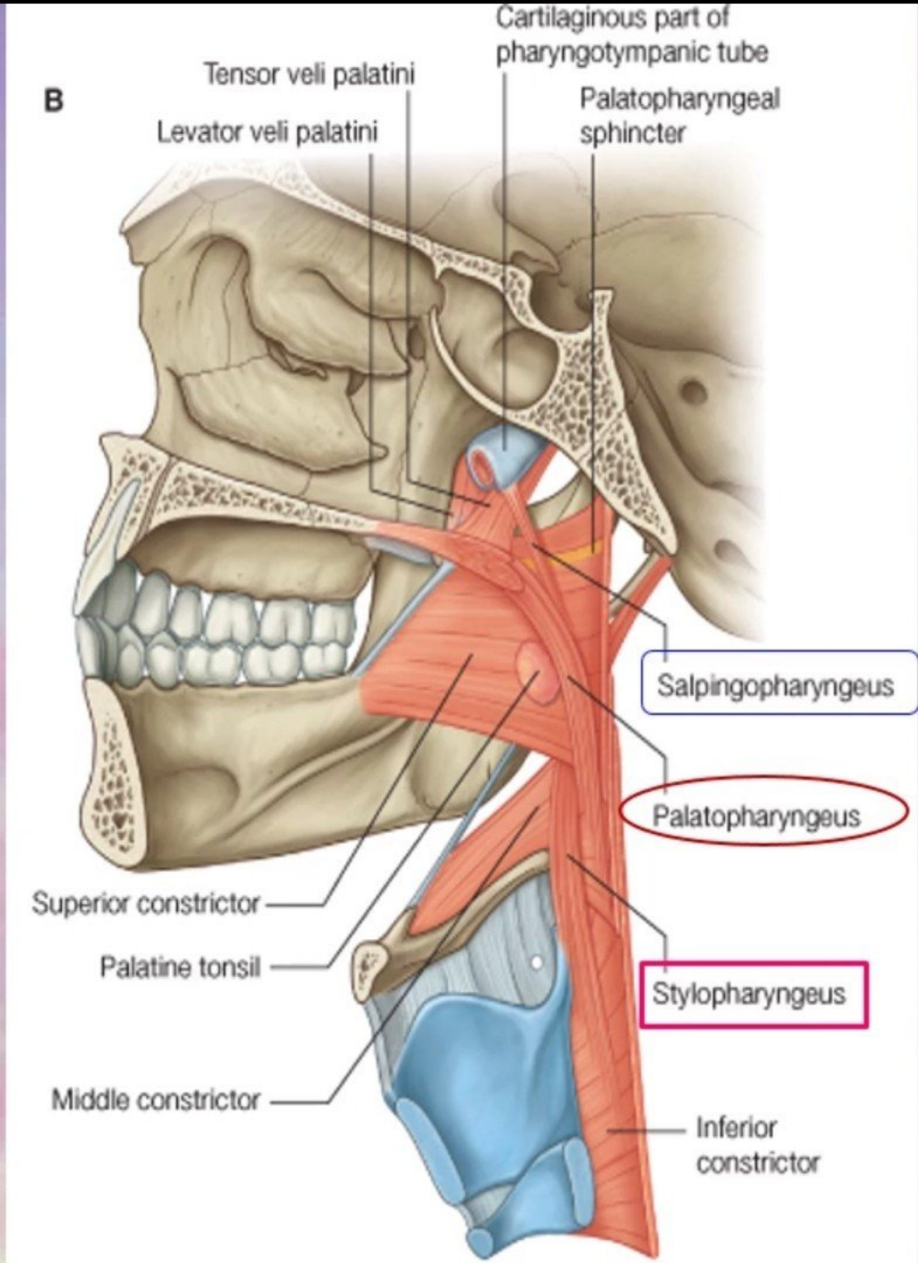


- Pharynx is divided into three parts:
  - **Nasopharynx.**
  - **Oropharynx.**
  - **Laryngopharynx.**



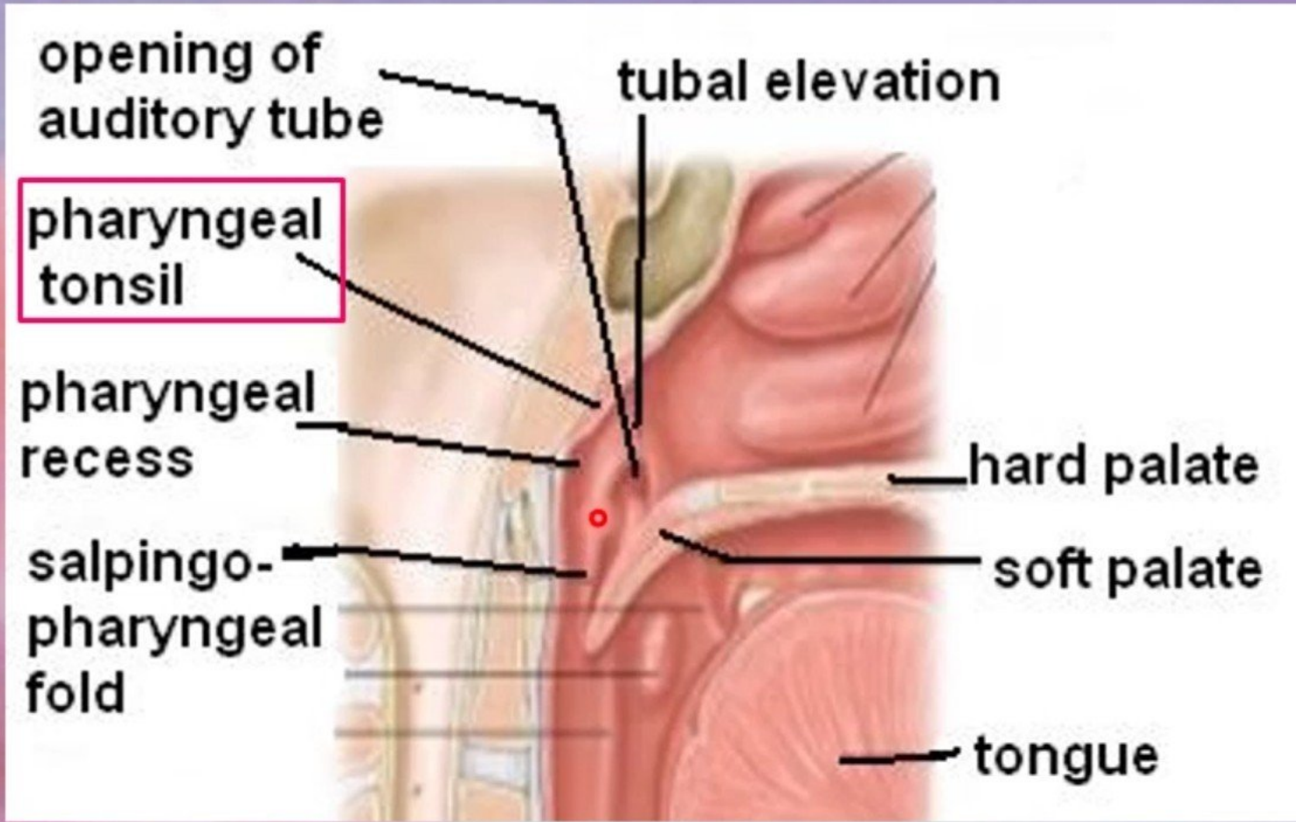
# Longitudinal Muscles

- **Three in number:**
  - **Stylopharyngeus**
  - **Salpingopharyngeus**
  - **Palatpharyngeous**
- **Function:**
  - Elevate the larynx & pharynx during swallowing.



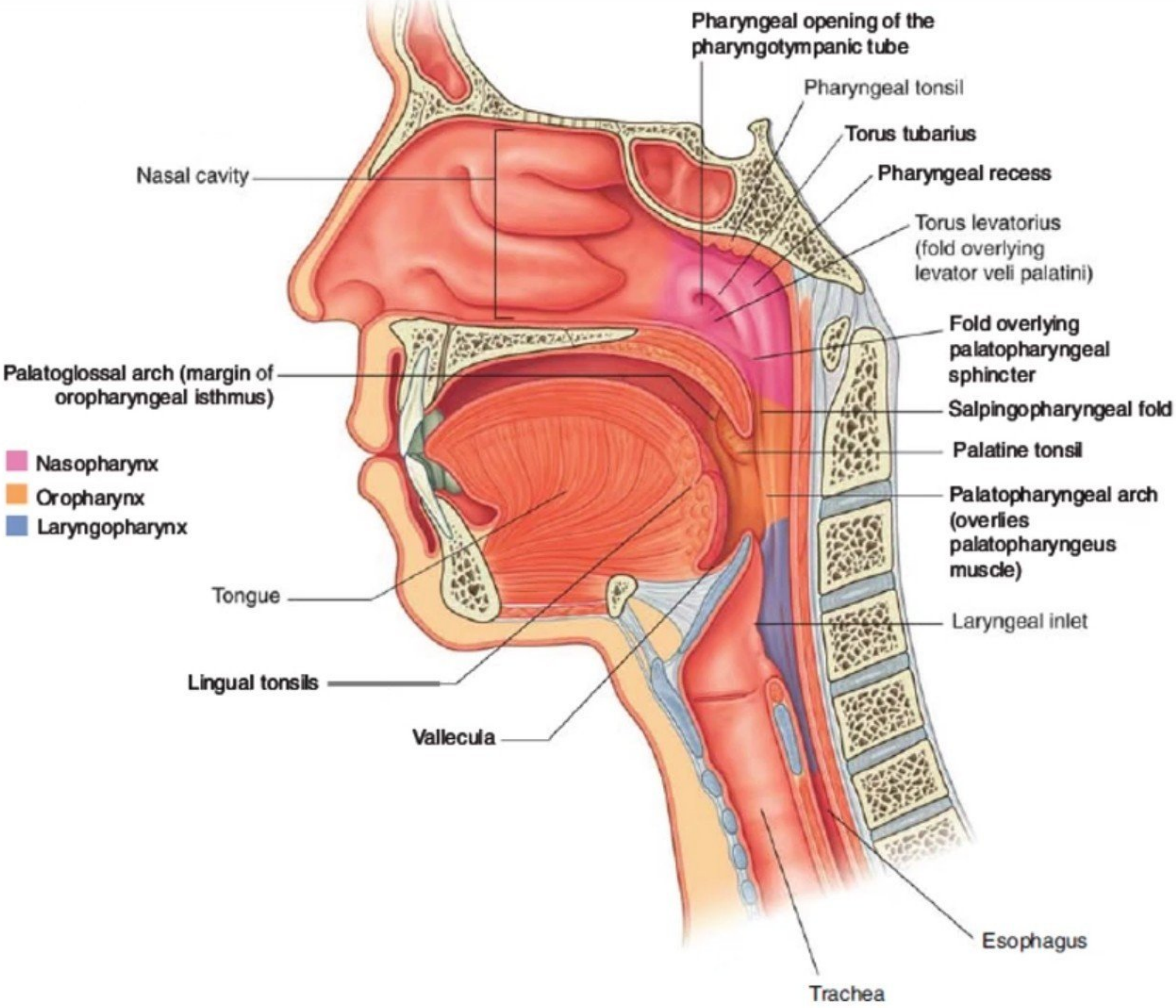
# Nasopharynx

- Extends from the base of skull to the soft palate.
- communicates with the nasal cavity through posterior nasal apertures.
- **Pharyngeal tonsils** (Adenoides) present in the submucosa covering the roof.
- **Lateral wall shows:**
  - **Opening of auditory tube**.
  - **Tubal elevation** (produced by posterior margin of the auditory tube).
  - **Tubal tonsil.**
  - **Pharyngeal recess.**
  - **Salpingopharyngeal fold** (raised by salpingo-pharyngeus muscle).



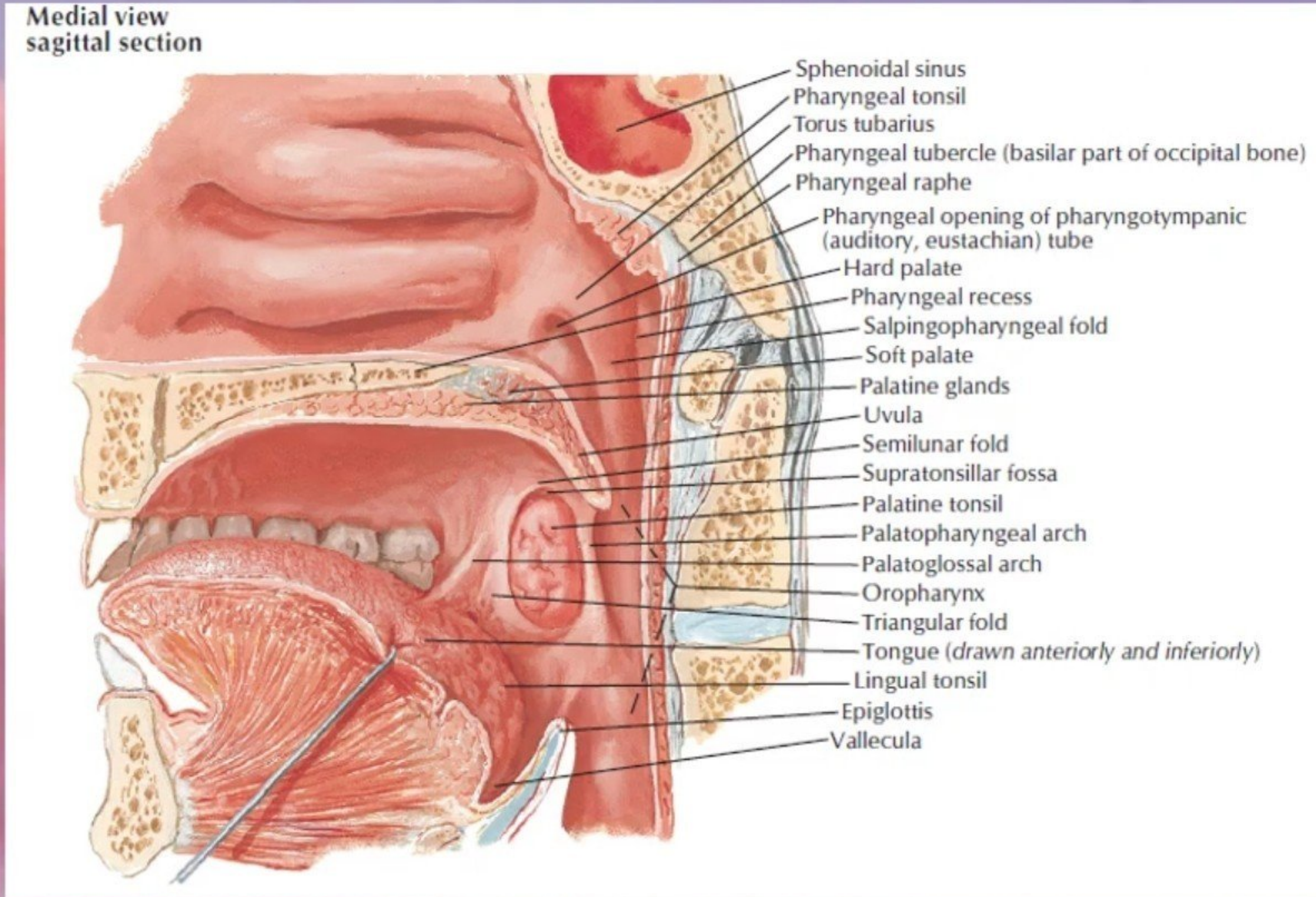
**The pharyngeal recess** (fossa of Rosenmüller):

- It is a narrow vertical gutter behind the opening of the auditory tube, resulting from the angular attachment of the pharyngobasilar fascia to the base of the skull in front of the carotid canal.
- A catheter missing the tubal orifice and introduced into the recess may perforate the fascia and enter the internal carotid artery, which here lies against the wall of the pharynx.



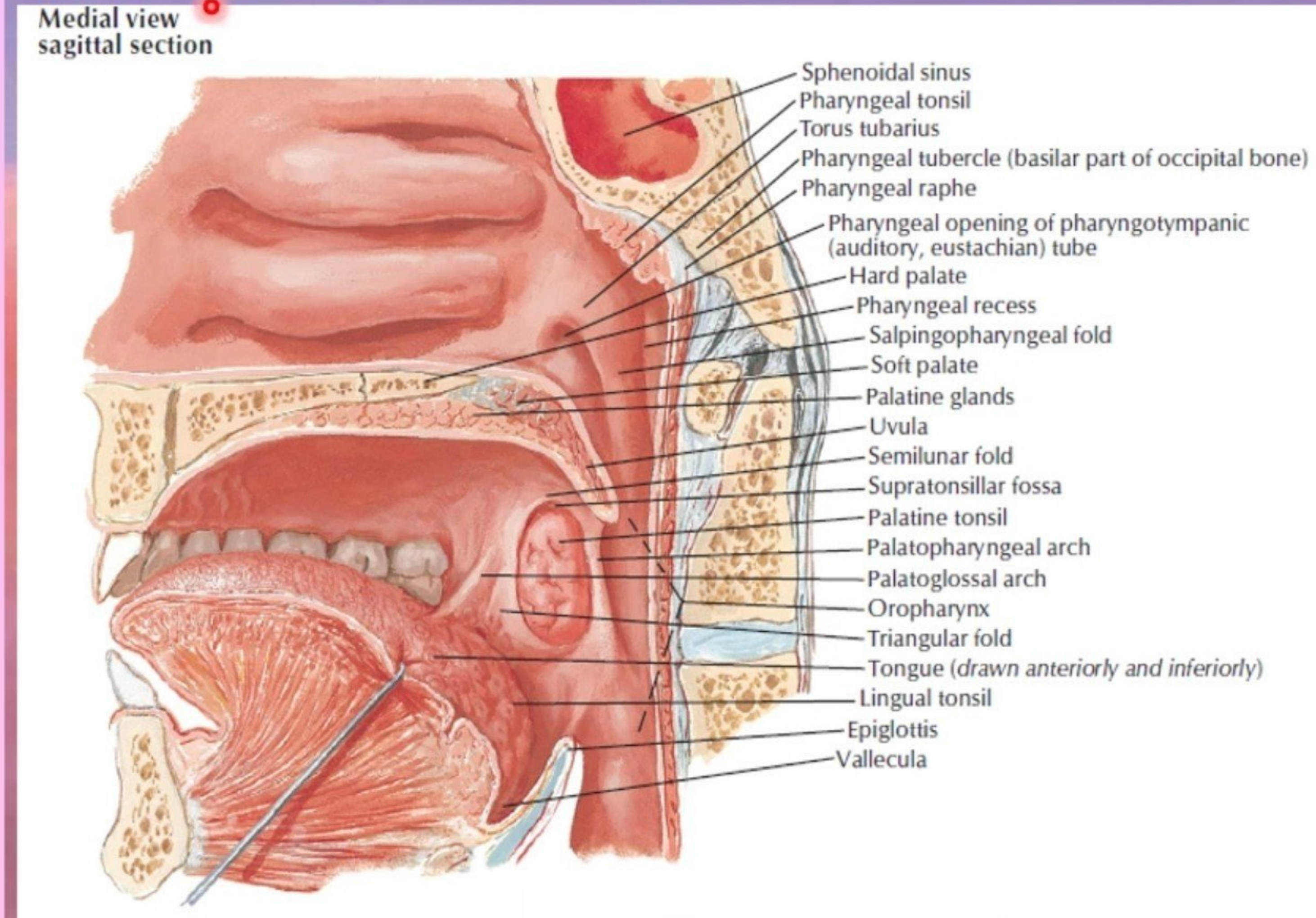
# Oropharynx

- Lies behind the mouth, communicates with the oral cavity through the **oropharyngeal isthmus**
- Extends from soft palate to upper border of epiglottis.
- **Lateral wall shows:**
  - **Palatopharyngeal fold.**
  - **Palatoglossal fold**
  - **Palatine tonsil** located between them in a depression called the **'tonsillar fossa'.**

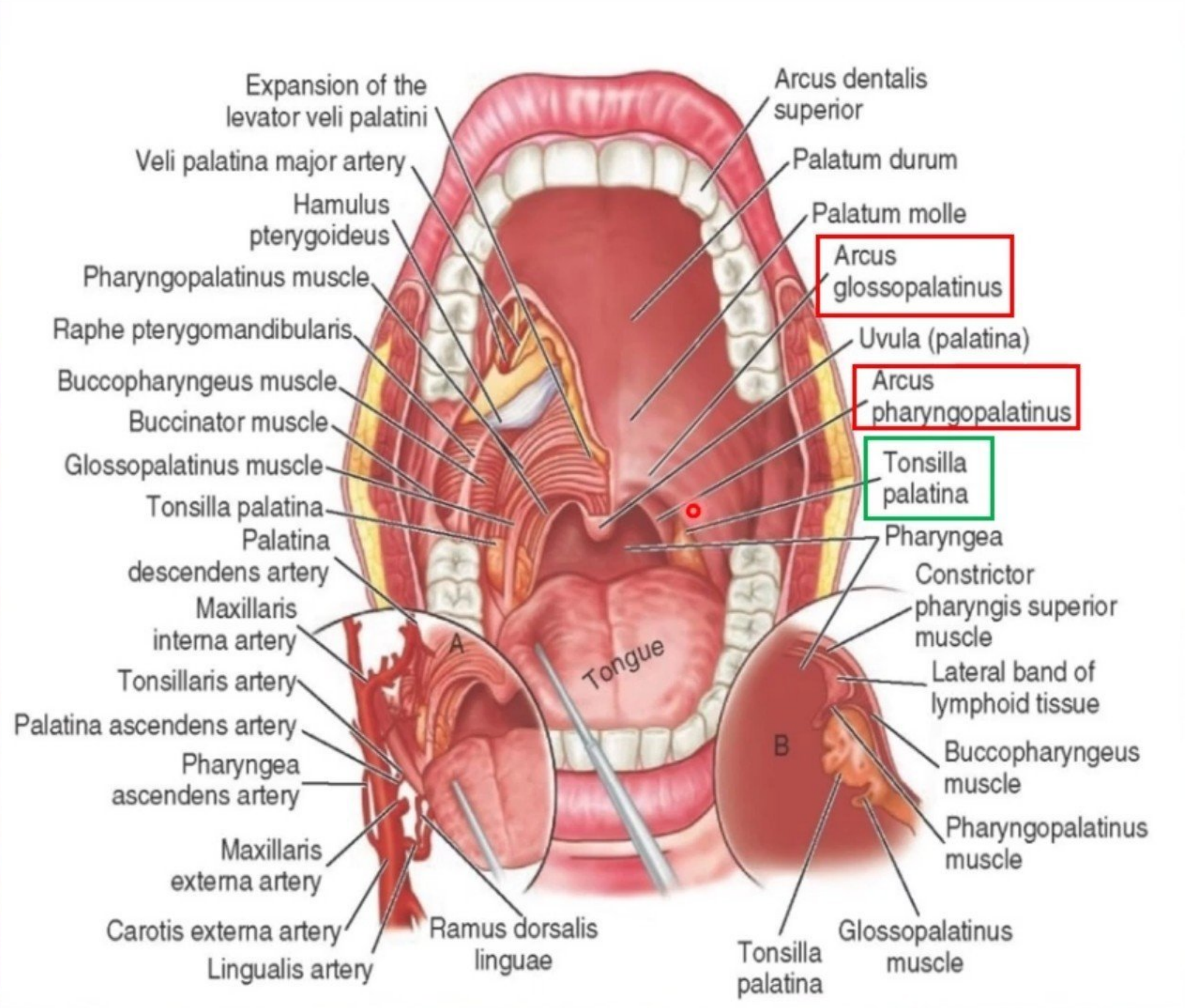


# The palatine tonsil (tonsils)

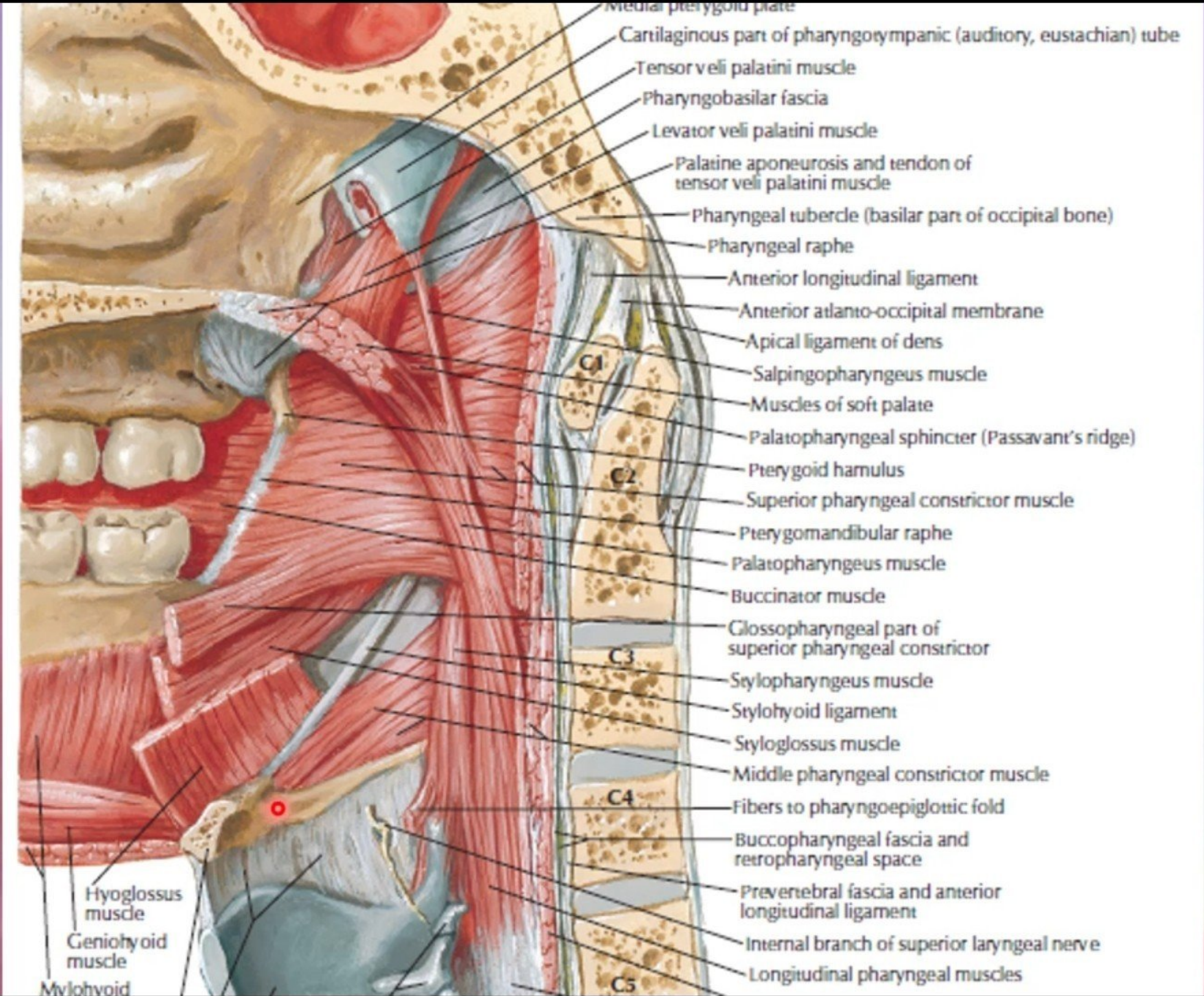
- It is a large collection of lymphoid tissue which projects into the oropharynx.
- It is found in the *tonsillar fossa* between the palatopharyngeal fold behind and the palatoglossal fold in front.
- The glossopharyngeal nerve crosses the lower part of the floor of the *tonsillar*.
- The superior constrictor separates this surface from the facial artery and two of its branches, the ascending palatine and tonsillar.




- The internal carotid artery is about 2.5 cm posterolateral to the tonsil.

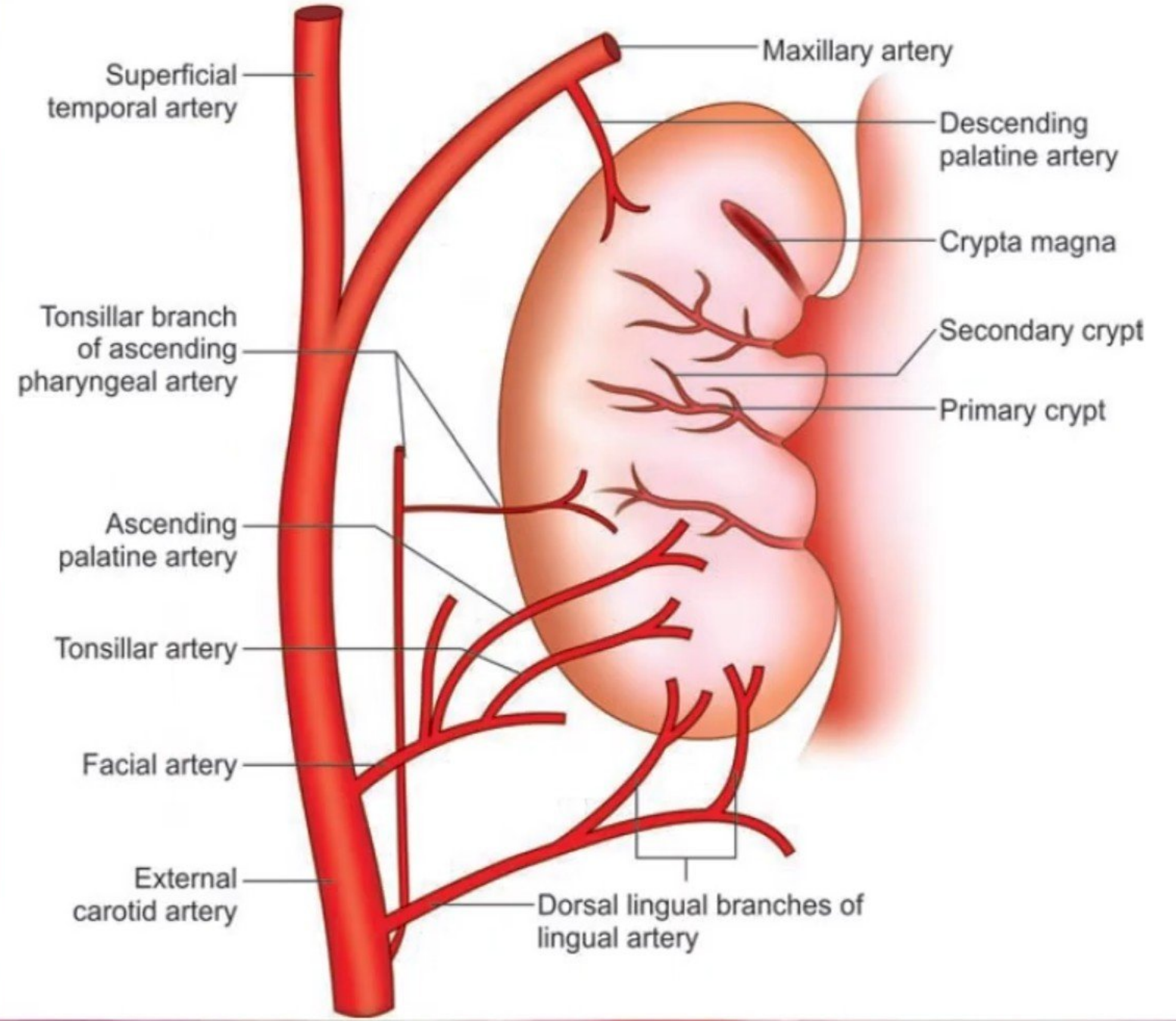


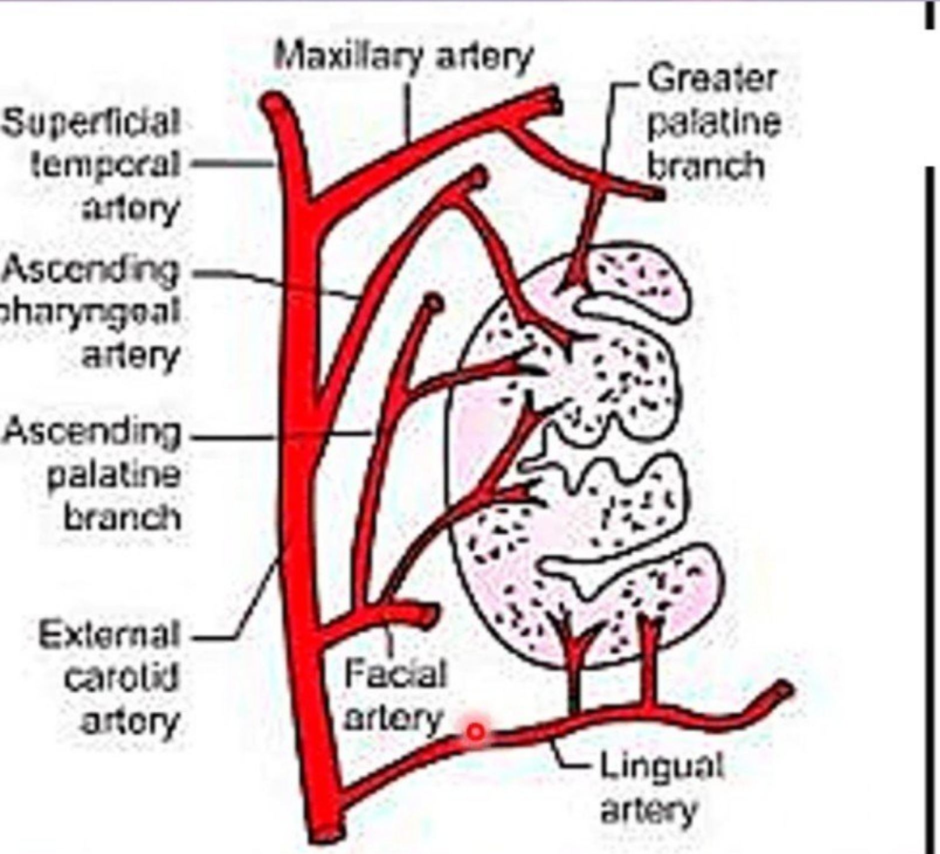
Expansion of the levator veli palatini
Veli palatina major artery
Hamulus pterygoideus
Pharyngopalatinus muscle
Raphe pterygomandibularis
Buccopharyngeus muscle
Buccinator muscle
Glossopalatinus muscle
Tonsilla palatina
Palatina descendens artery
Maxillaris interna artery
Tonsillaris artery
Palatina ascendens artery
Pharyngea ascendens artery
Maxillaris externa artery
Carotis externa artery
Lingualis artery
Ramus dorsalis linguae
Arcus dentalis superior
Palatum durum
Palatum molle
Arcus glossopalatinus
Uvula (palatina)
Arcus pharyngopalatinus
Tonsilla palatina
Pharyngea
Constrictor pharyngis superior muscle
Lateral band of lymphoid tissue
Buccopharyngeus muscle
Pharyngopalatinus muscle
Glossopalatinus muscle
Tonsilla palatina
Tongue
A
B


Medial pterygoid plate
Cartilaginous part of pharyngotympanic (auditory, eustachian) tube
Tensor veli palatini muscle
Pharyngobasilar fascia
Levator veli palatini muscle
Palatine aponeurosis and tendon of tensor veli palatini muscle
Pharyngeal tubercle (basilar part of occipital bone)
Pharyngeal raphe
Anterior longitudinal ligament
Anterior atlanto-occipital membrane
Apical ligament of dens
Salpingopharyngeus muscle
Muscles of soft palate
Palatopharyngeal sphincter (Passavant's ridge)
Pterygoid hamulus
Superior pharyngeal constrictor muscle
Pterygomandibular raphe
Palatopharyngeus muscle
Buccinator muscle
Glossopharyngeal part of superior pharyngeal constrictor
Stylopharyngeus muscle
Stylohyoid ligament
Styloglossus muscle
Middle pharyngeal constrictor muscle
Fibers to pharyngoepiglottic fold
Buccopharyngeal fascia and retropharyngeal space
Prevertebral fascia and anterior longitudinal ligament
Internal branch of superior laryngeal nerve
Longitudinal pharyngeal muscles
C1
C2
C3
C4
C5
Hyoglossus muscle
Geniohyoid muscle
Mylohyoid

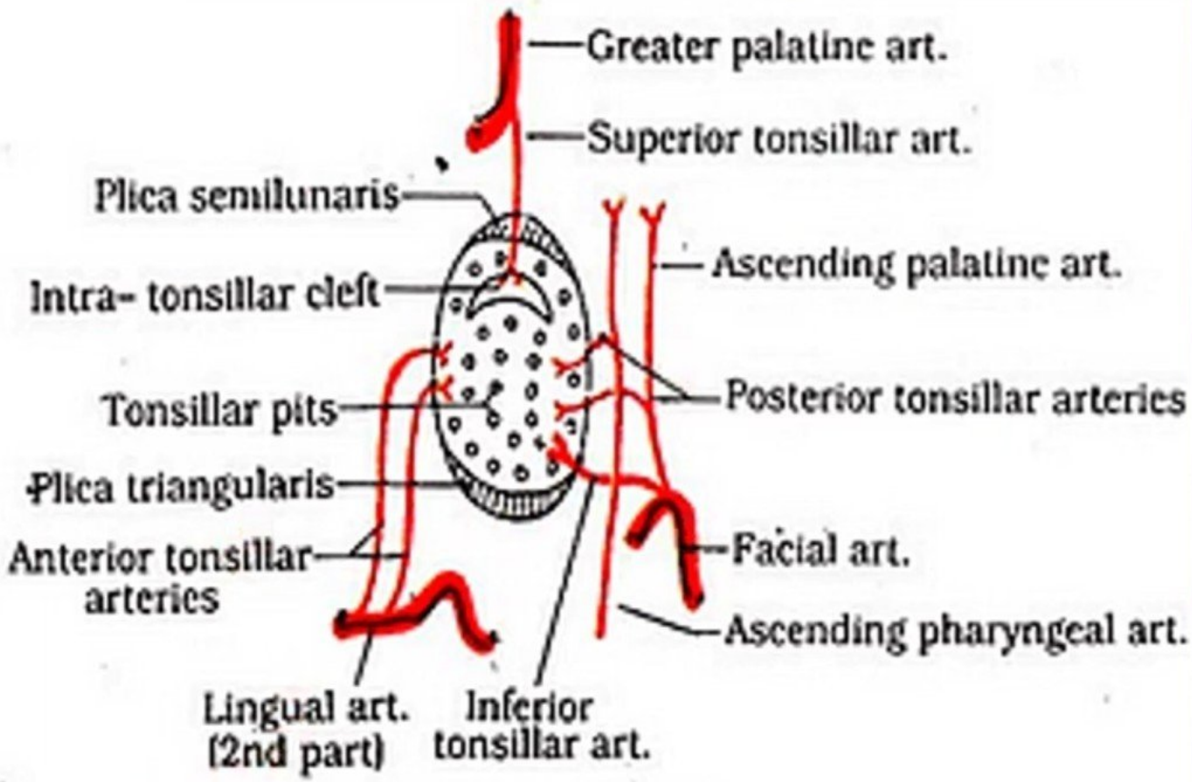
## Blood supply:

- The tonsillar branch of the facial artery forms the main arterial supply.
- It enters the tonsil by piercing the superior constrictor.
- There are smaller contributions from the lingual, ascending pharyngeal and ascending and greater palatine vessels.





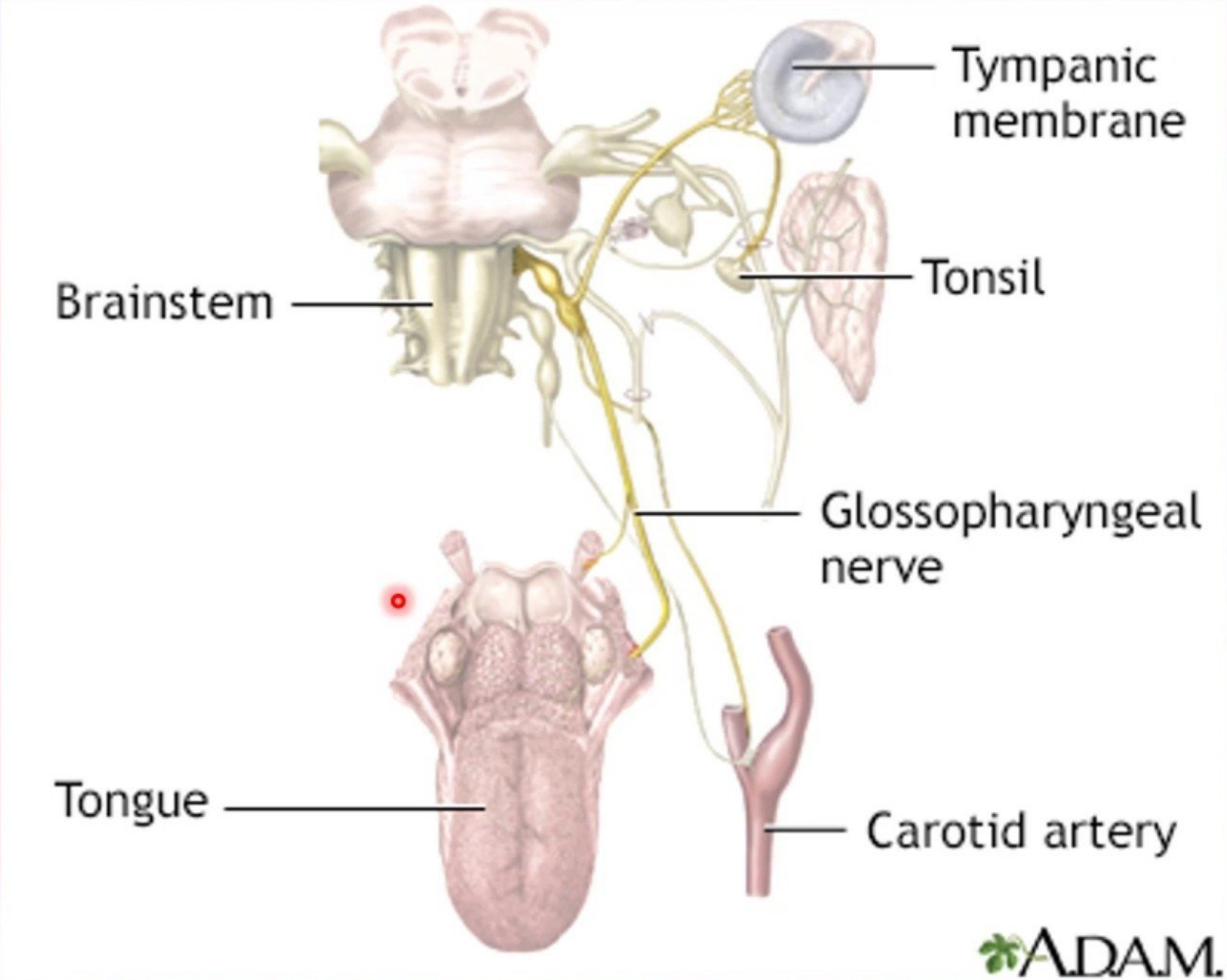
Maxillary artery
Greater palatine branch
Superficial temporal artery
Ascending pharyngeal artery
Ascending palatine branch
External carotid artery
Facial artery
Lingual artery

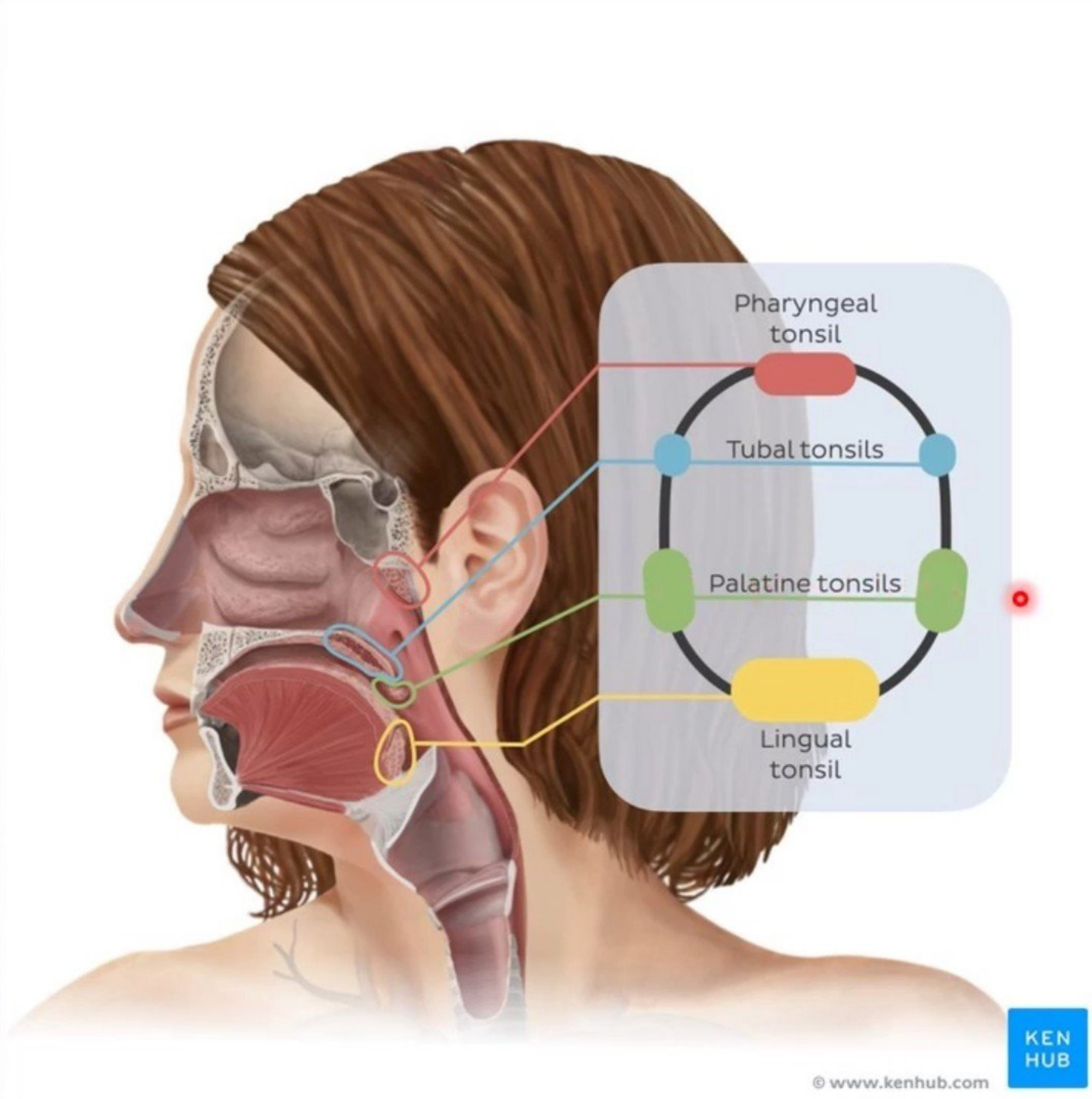
Greater palatine art.
Superior tonsillar art.
Plica semilunaris
Intra- tonsillar cleft
Ascending palatine art.
Tonsillar pits
Posterior tonsillar arteries
Plica triangularis
Anterior tonsillar arteries
Facial art.
Ascending pharyngeal art.
Lingual art. (2nd part)
Inferior tonsillar art.

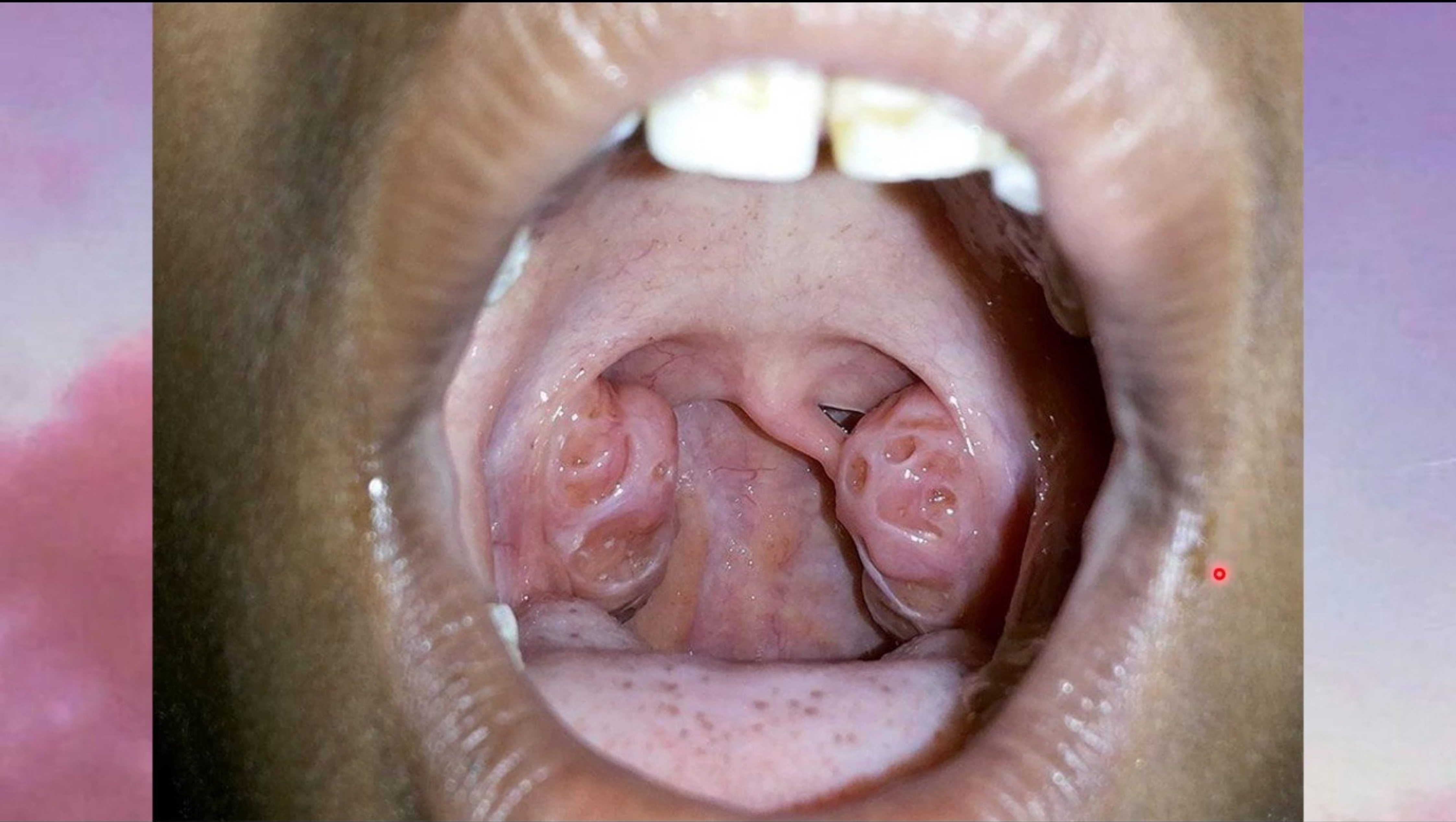
**Nerve supply.**

- The mucous membrane overlying the tonsil is supplied mainly by the tonsillar branch of the glossopharyngeal nerve, and to a small extent by the lesser palatine nerves.
- The glossopharyngeal nerve also supplies the middle ear, through its tympanic branch, and tonsillitis may cause referred pain in the ear.



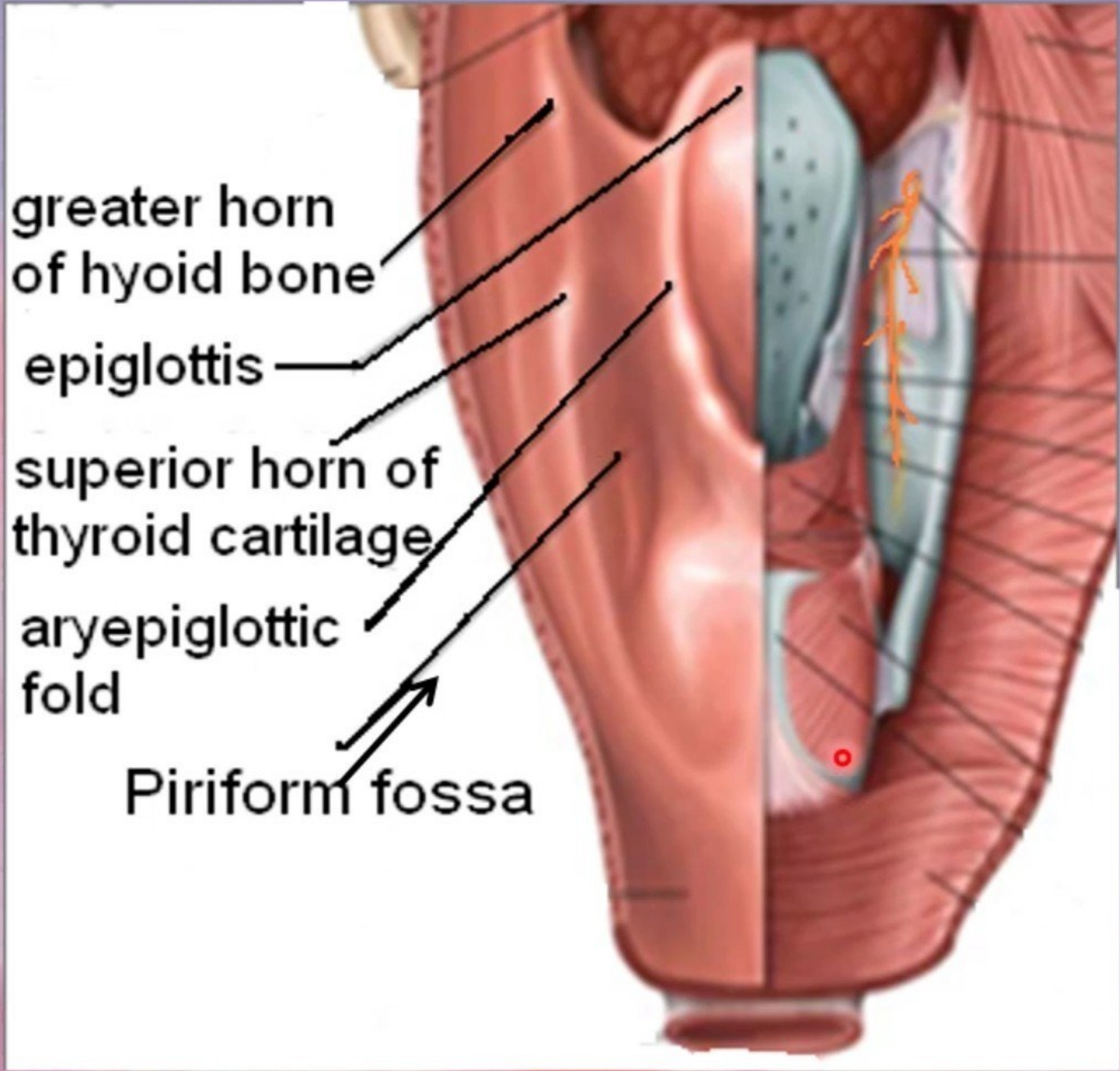
The **palatine**, **lingual**, **pharyngeal** and **tubal tonsils** collectively form an interrupted circle of **lymphoid tissue** (***Waldeyer's ring***) at the upper end of the respiratory and alimentary tracts.





# Laryngopharynx

- Lies behind the laryngeal inlet & the posterior surface of larynx.
- communicates with the larynx through the laryngeal inlet
- Extends from upper border of epiglottis to lower border of cricoid cartilage.
- A small depression situated on either side of the laryngeal inlet is called '**Piriform Fossa**'.
- It is a common site for the lodging of foreign bodies.
- Branches of internal laryngeal & recurrent laryngeal nerves lie deep to the mucous membrane of the fossa and are vulnerable to injury during removal of a foreign body.



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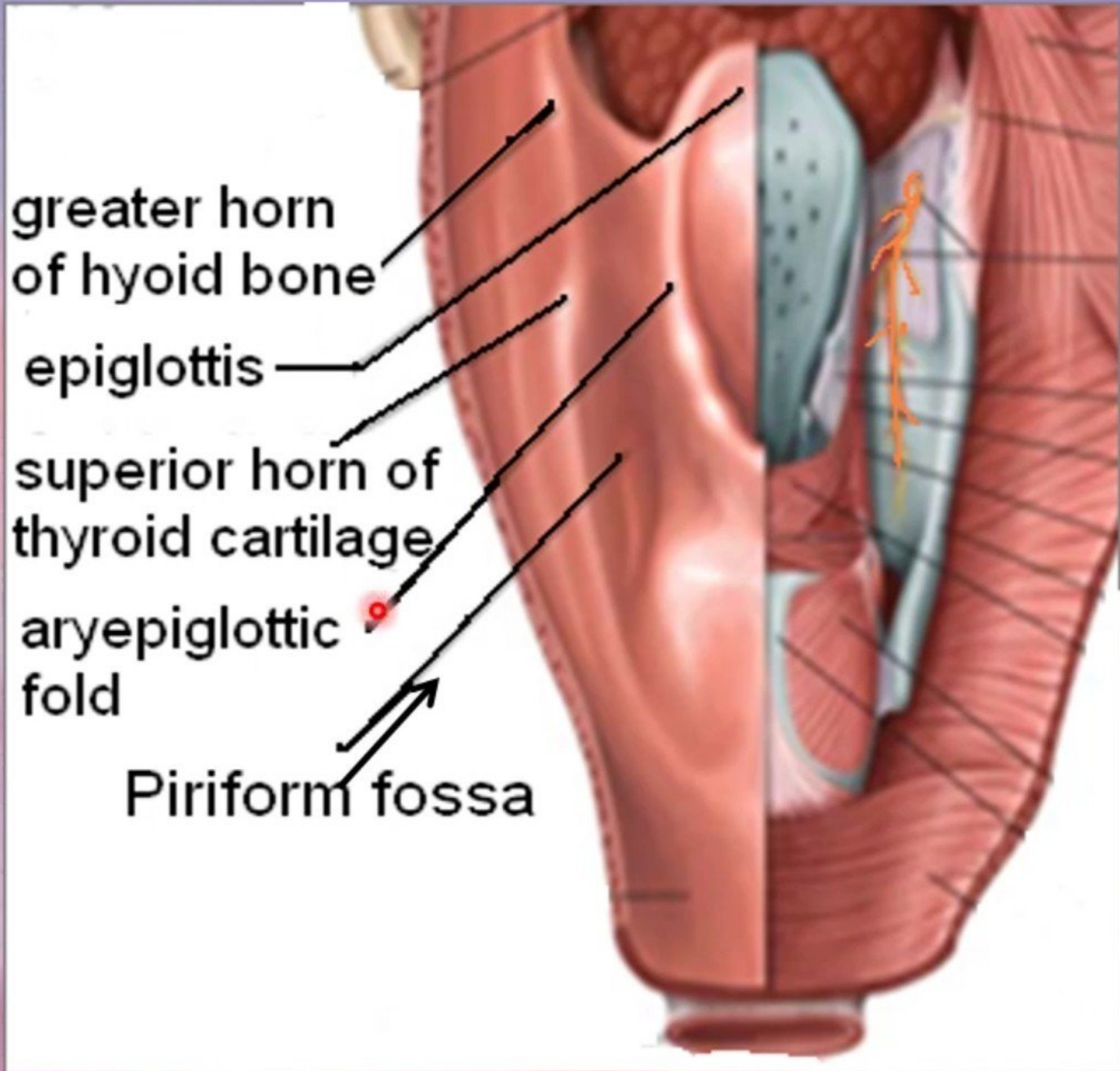
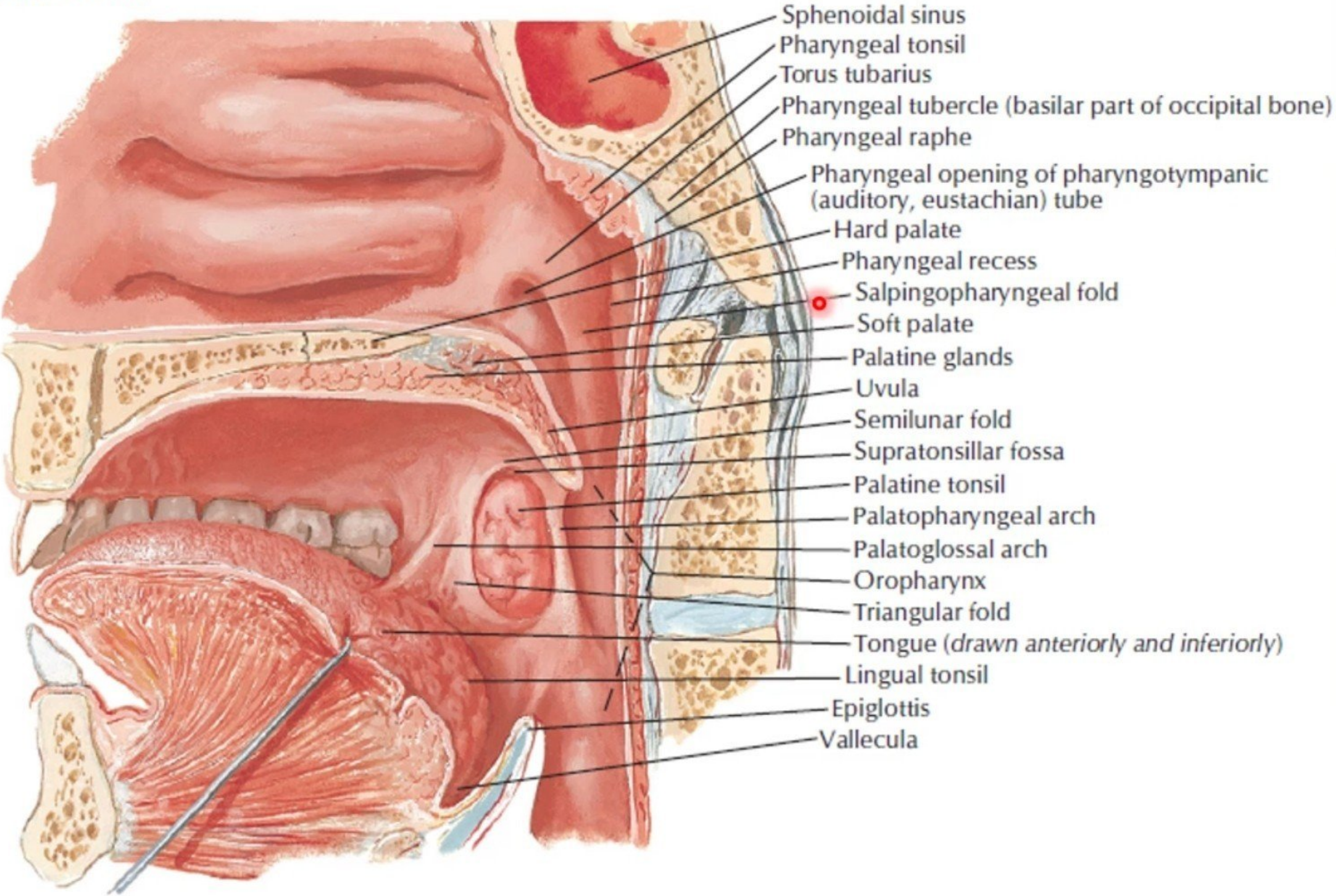
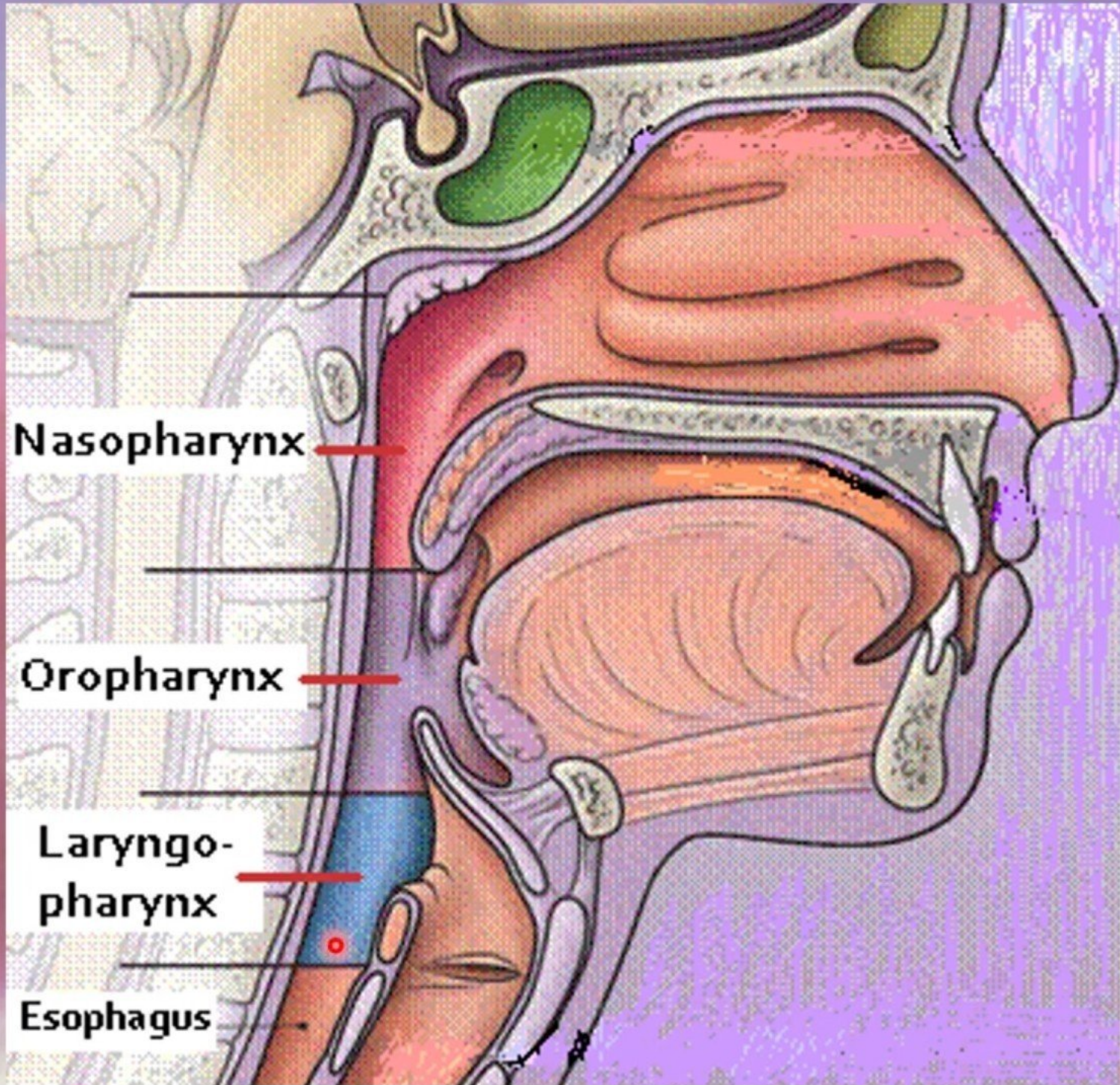


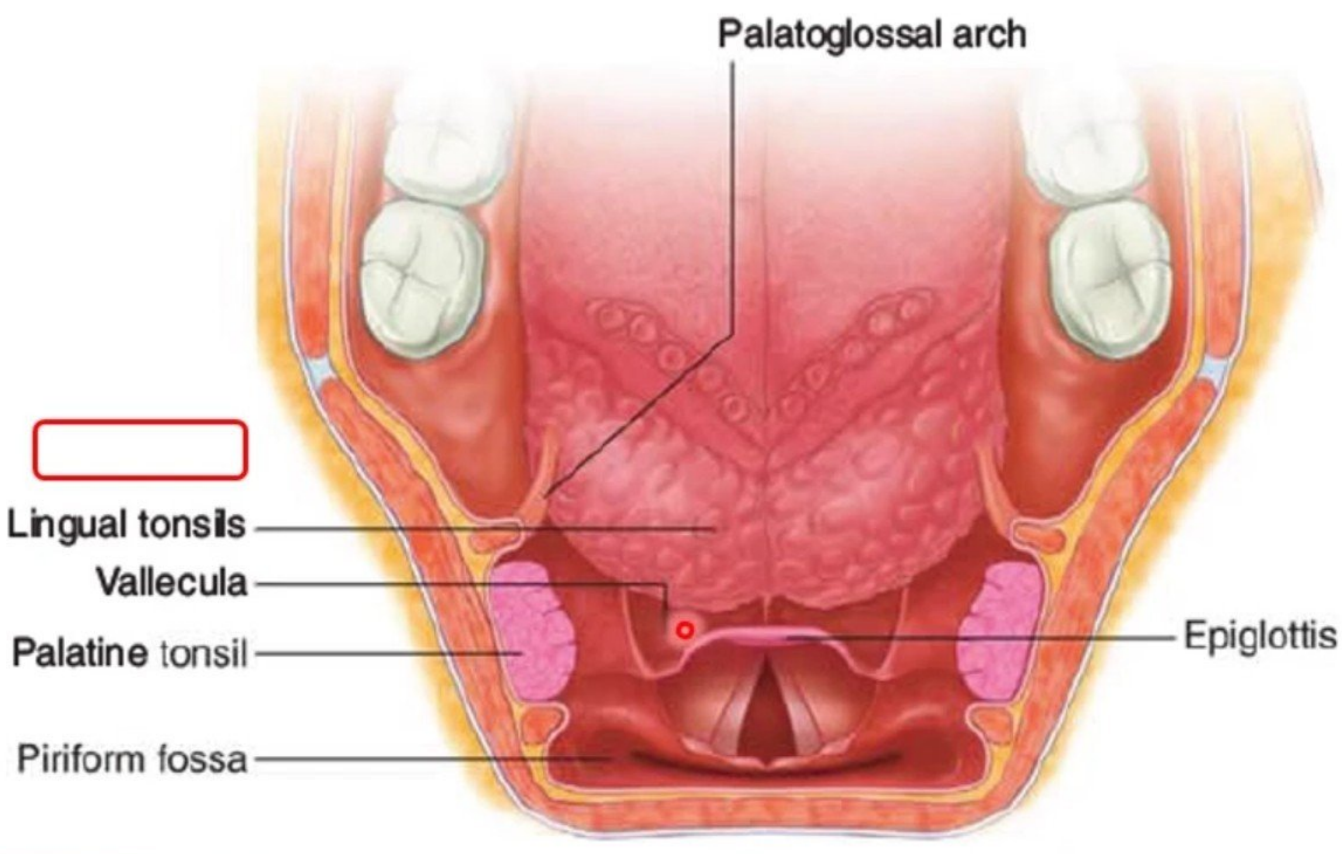
Medial view
sagittal section




Nasopharynx
Oropharynx
Laryngo-
pharynx
Esophagus

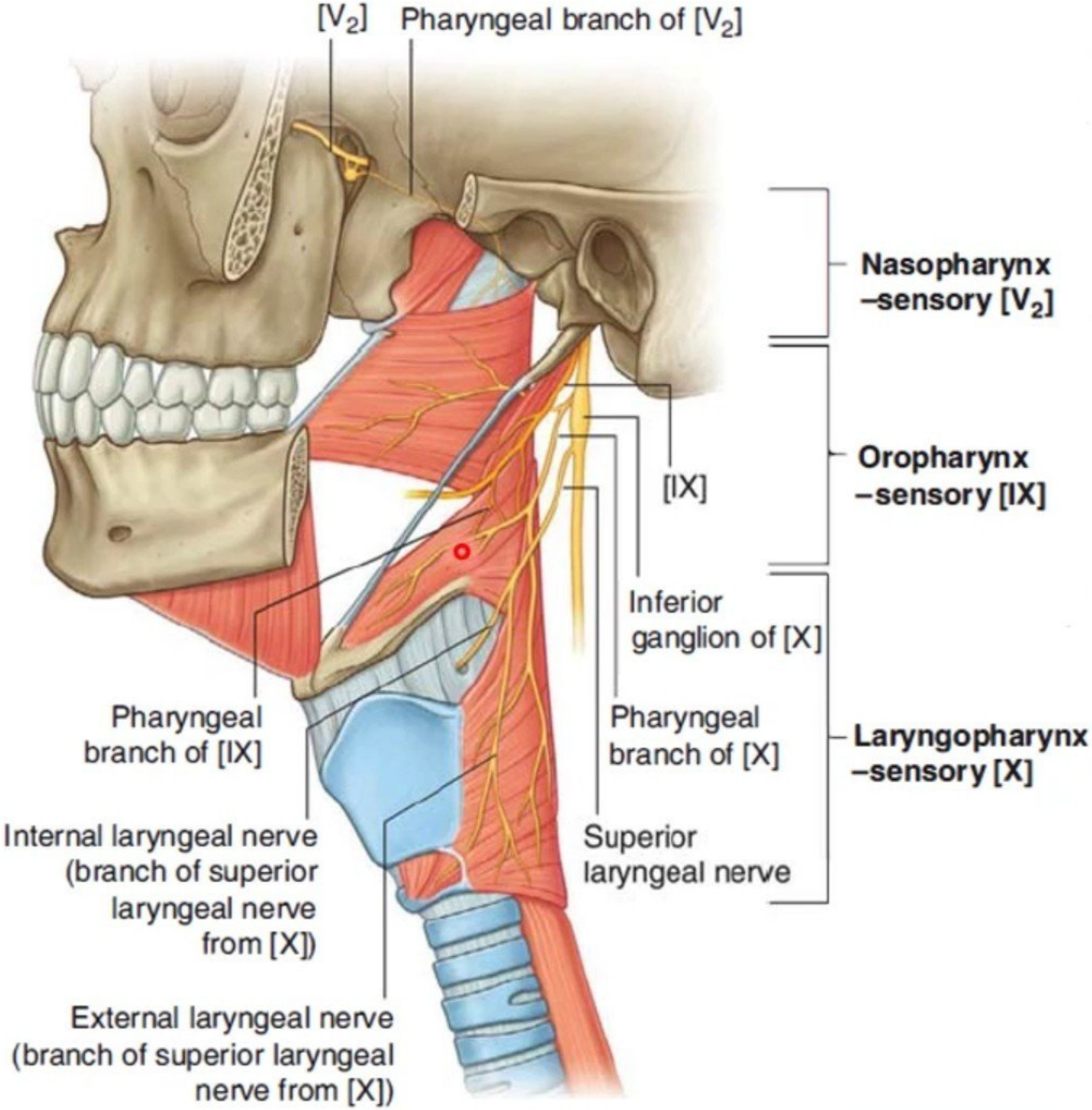
# The valleculae

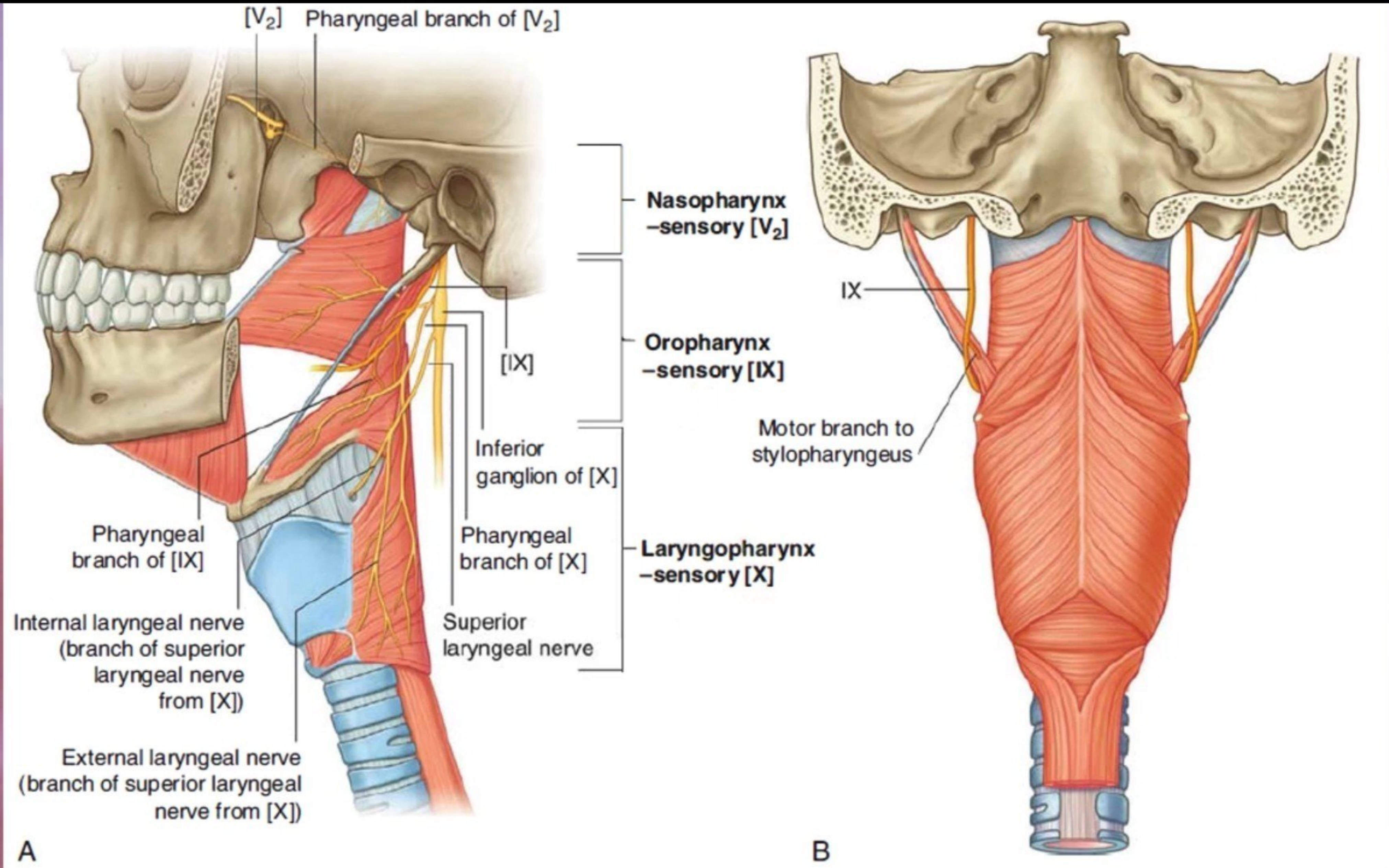
- They are lie between the epiglottis and the posterior surface of the tongue.
- They are shallow fossae separated by the median glossoepiglottic fold and limited inferolaterally by the lateral glossoepiglottic folds.
- The nerve supply of the mucosa by the internal laryngeal nerve.
- If any thing that lodges in the vallecula that stimulate a coughing reflex to dislodge it.



## Nerve Supply

- **Sensory: (mucosa)**
- **Nasopharynx:** Maxillary nerve
  (supplied by the pharyngeal branch of the maxillary nerve through the pterygopalatine ganglion)
- **Oropharynx:** Glossopharyngeal nerve
  (but the internal laryngeal nerve supplies the valleculae and the lesser palatine nerves (maxillary) contribute to the supply of the tonsillar mucosa)
- **Laryngopharynx:** Vagus nerve
  (The internal laryngeal nerve)
- **Motor :**
  - All the muscles of pharynx are supplied by the pharyngeal plexus.
  - Except ; the Stylopharyngeus is supplied by the glossopharyngeal nerve.
- **Cricopharyngeus** is also supplied by the recurrent and external laryngeal nerves.

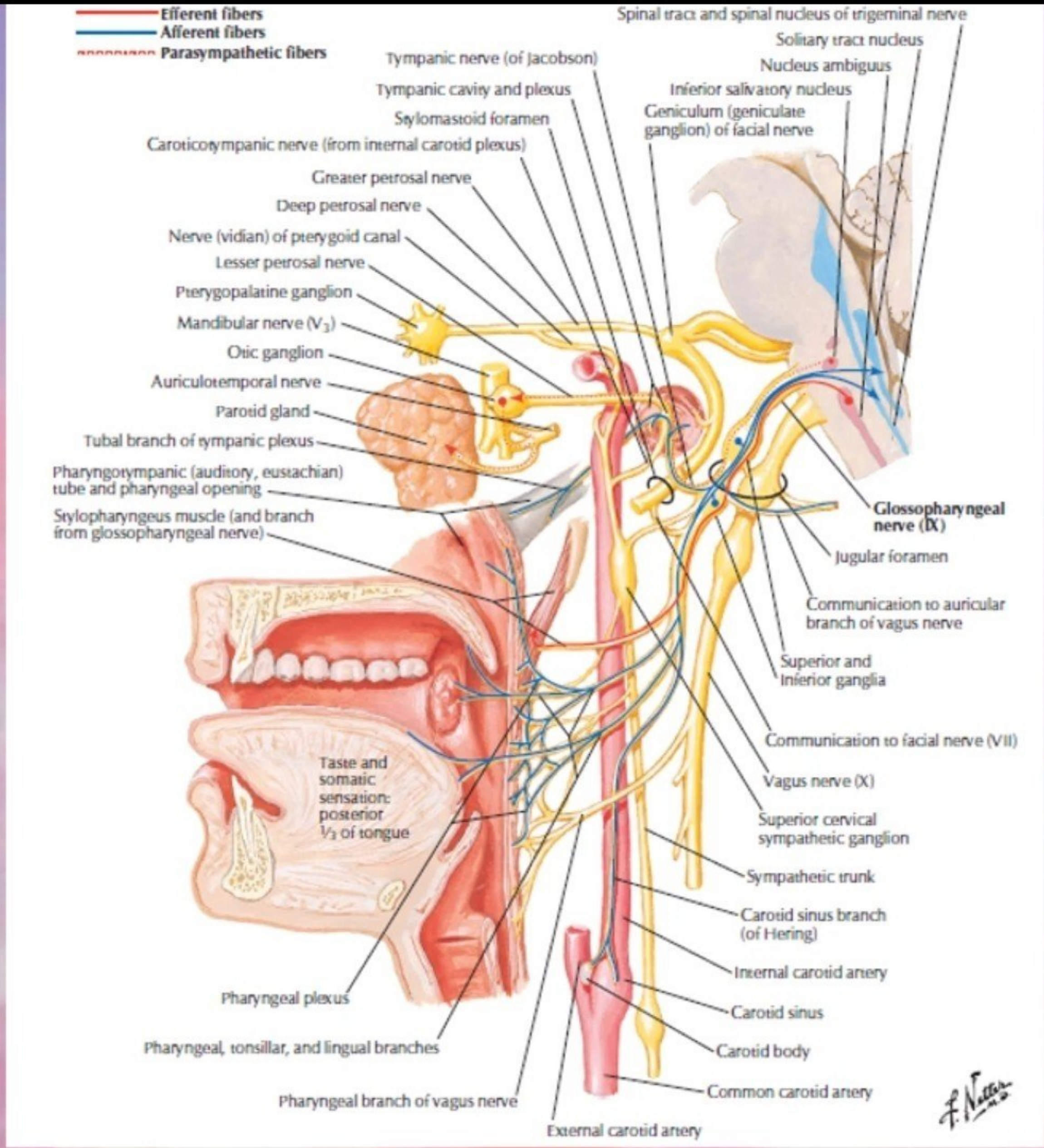




[V2]
Pharyngeal branch of [V2]
Nasopharynx –sensory [V2]
Oropharynx –sensory [IX]
[IX]
Inferior ganglion of [X]
Laryngopharynx –sensory [X]
Pharyngeal branch of [IX]
Pharyngeal branch of [X]
Internal laryngeal nerve (branch of superior laryngeal nerve from [X])
Superior laryngeal nerve
External laryngeal nerve (branch of superior laryngeal nerve from [X])
A
IX
Motor branch to stylopharyngeus
B

# The pharyngeal plexus

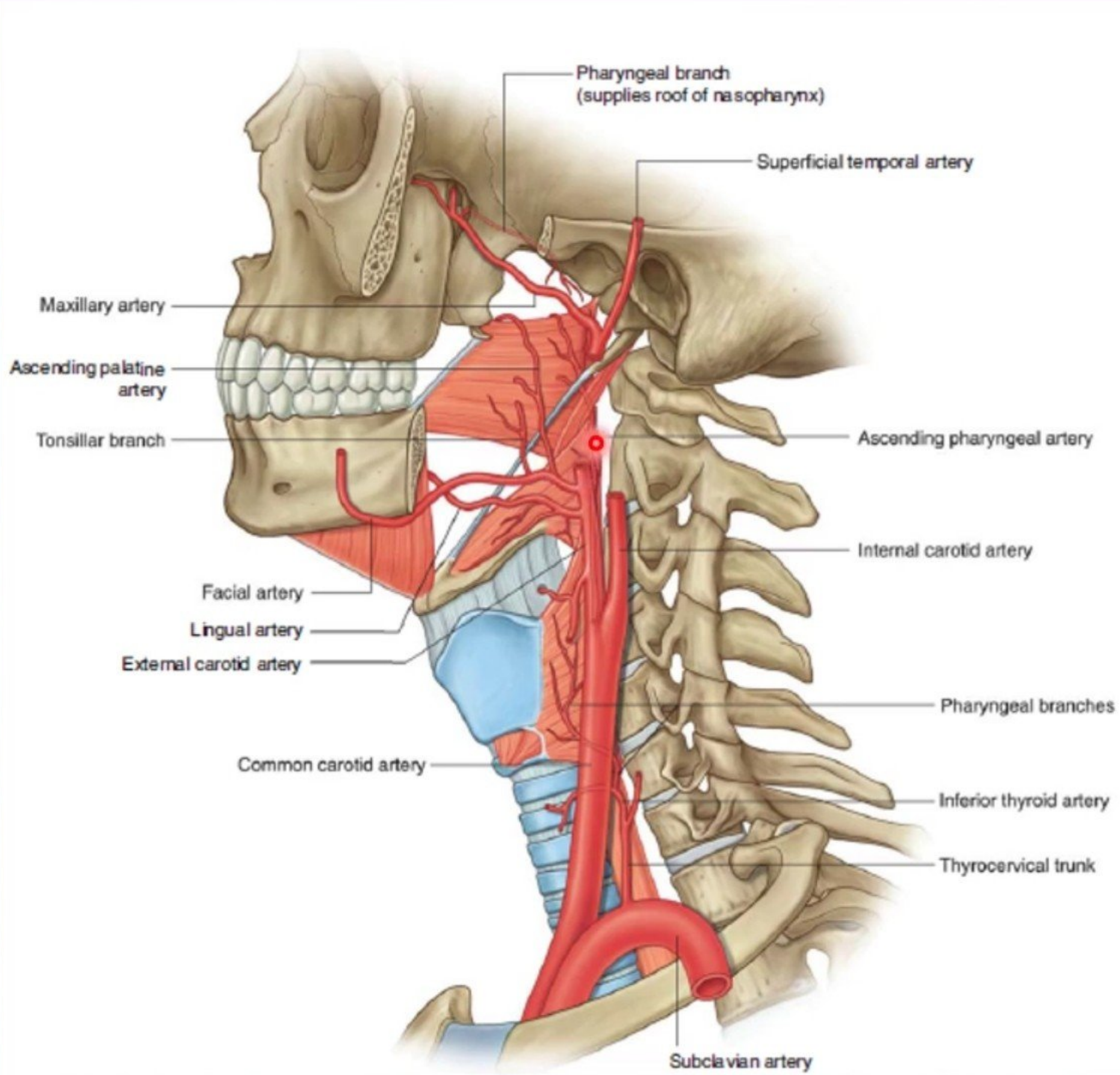
- It is lies on the posterolateral wall of the pharynx, mainly over the middle constrictor.
- It is formed by the union of:
    - The pharyngeal branches from the vagus.
    - The pharyngeal branches from glossopharyngeal nerves and
    - The external laryngeal nerve branches from the superior laryngeal branch of the vagus nerve.
    - the cervical sympathetic nerves.
- The glossopharyngeal component is afferent.
- the pharyngeal fibres of the vagus carry motor fibres (derived from the cranial part of the accessory nerve).
- The sympathetic fibres are vasoconstrictor.
- **<u>Note:</u>**
    - The pharyngeal branches from the vagus is the major motor nerve supply of the pharynx.
    - All muscles of the pharynx are innervated by vagus nerve (through the pharyngeal plexus) except ; the Stylopharyngeus is supplied by the glossopharyngeal nerve.



# Arterial supply:

from branches of the following arteries:

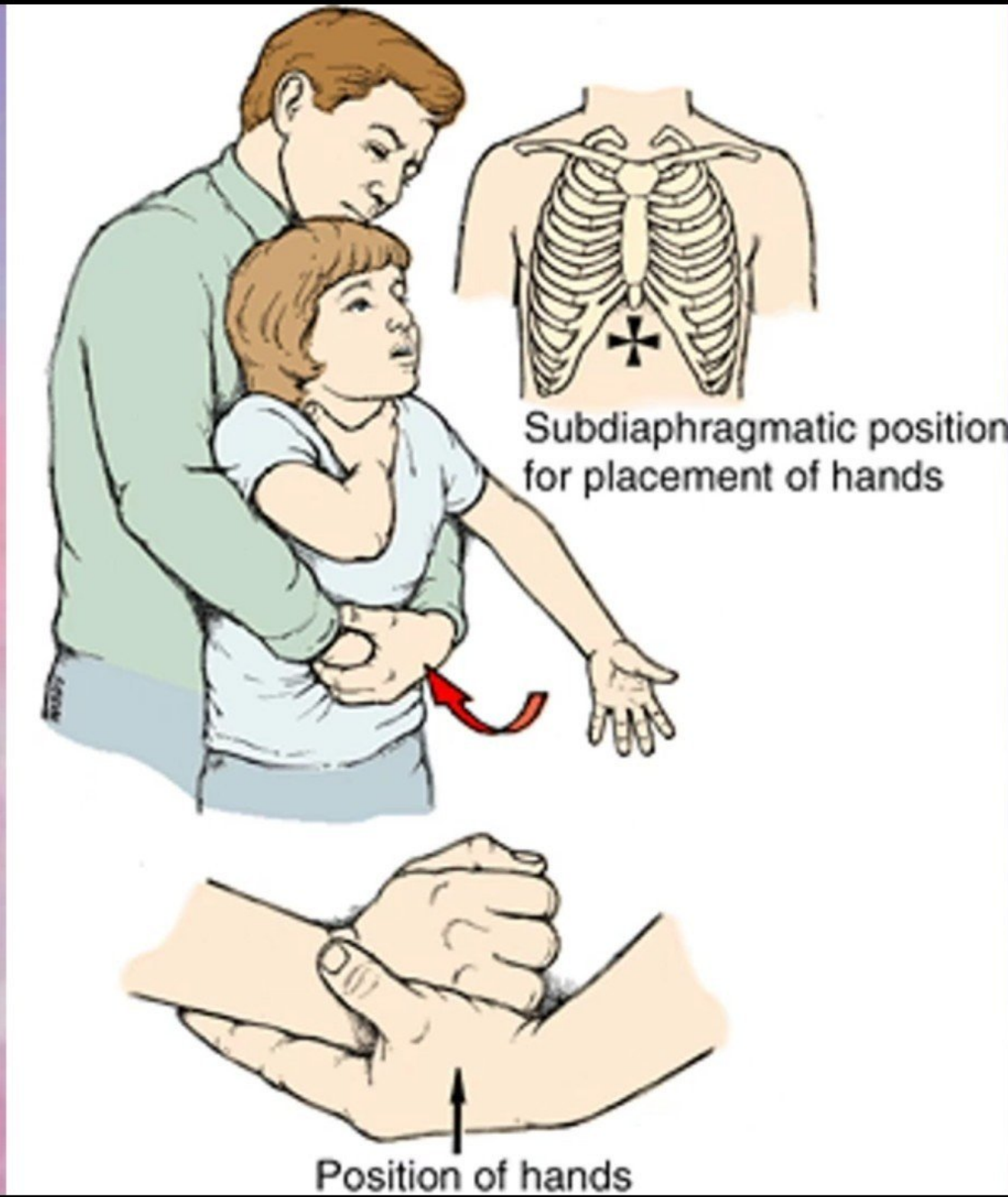
1) Facial artery supply pharynx by:
   - Ascending pharyngeal artery.
   - Ascending palatine artery.
   - Tonsillar artery.
2) Maxillary artery supply pharynx by:
   - greater palatine artery.
   - Pharyngeal artery.
3) lingual artery.
4) Superior and inferior laryngeal arteries.
5) Pharyngeal artery (from inferior thyroid artery).

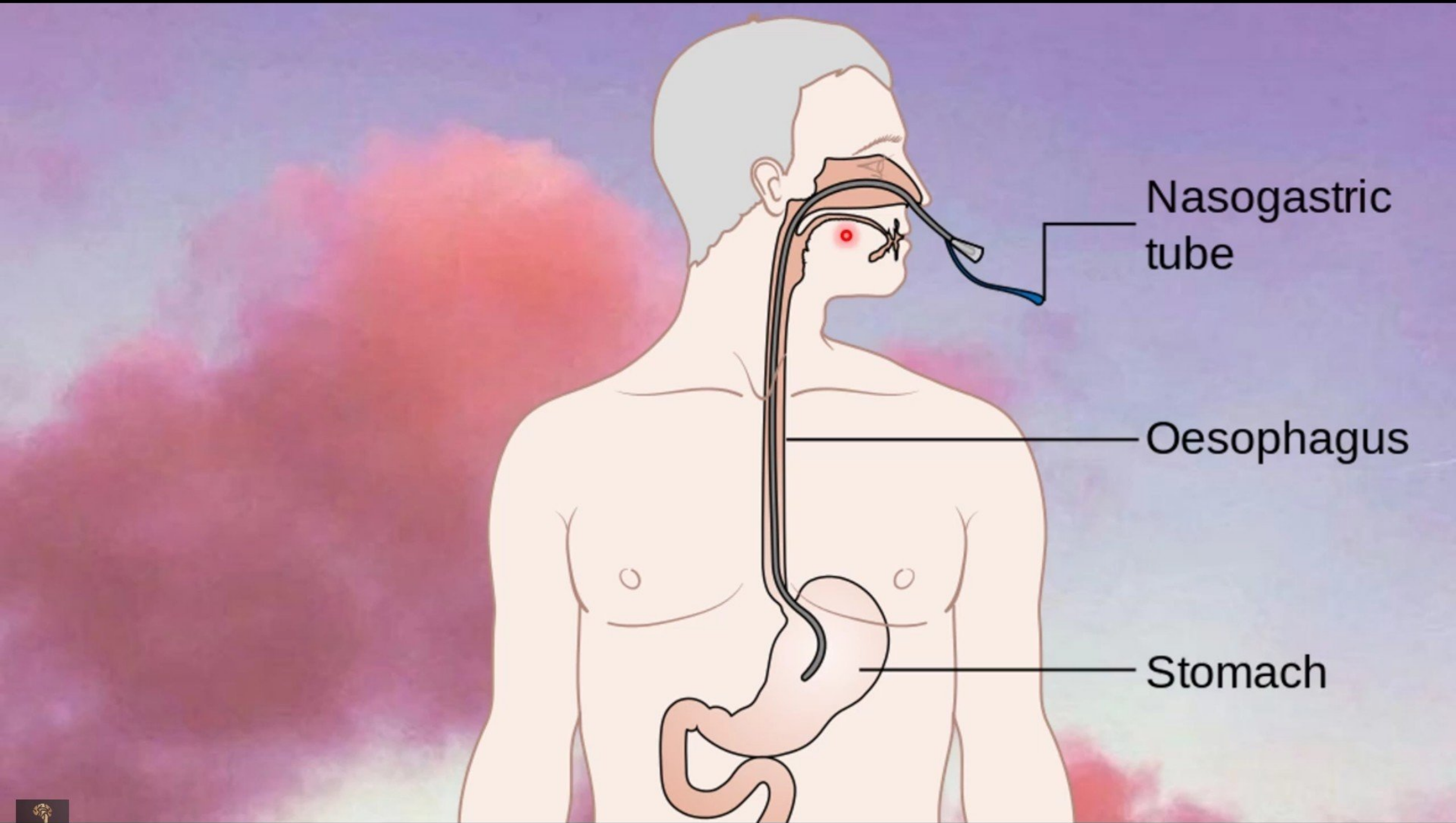


## Foreign Bodies in the Laryngopharynx

## Sinus Tract from the Piriform Recess

# Heimlich maneuver




Nasogastric
tube
Oesophagus
Stomach

# Thank You
# & Good Luck